Results Overview

• Designs generated: 60

• Designs passing all filters: 10

• MMCIF files of final designs are in:

workbench/real_run/final_ranked_designs/final_3_designs

• Metrics and sequences of 3 final designs are in:

workbench/real_run/final_ranked_designs/final_designs_metrics_3.csv

• Metrics of all designs are in:

workbench/real_run/final_ranked_designs/all_designs_metrics.csv

You can rerun filtering (very quick), using this command with changed parameters:

```
-- boltzgen run input_spec.yaml --steps filtering --config filtering budget=60 alpha=0.05
```

You can also rerun filtering in a jupyter notebook if you want using `filter.ipynb`

What was run to produce this in the Filter task:

1. Filtering: each design is evaluated against mandatory thresholds.
2. Ranking: for every metric we compute its rank, then scale it by the metric's inverse-importance weight.

Designs with fewer passed filters are automatically penalised because the ranking key is the pair (num_filters_passed, metric). The overall quality score is the worst (maximum) of these scaled ranks.

The 10 best designs form the Top set.

3. Diversity: a lazy-greedy algorithm selects 3 designs that jointly maximise quality and minimise sequence similarity (sequence-identity distance). The trade-off is controlled by α = 0.001:

• α = 0 → 100 % quality focus (same as Top set)

• α = 1 → 100 % diversity focus (ignores quality)

• Quality – composite of metrics such as iPTM, salt-bridges, ΔSASA, etc. Each metric has an "inverse importance" weight (see "Sorting Criteria" table). A larger weight divides the rank by a bigger number and therefore down-weights that metric.

• Diversity – 1 − sequence identity between designs.

We use α = 0.001, meaning 100 % emphasis on quality, 0 % on diversity.

CSV Column Reference

Column	Description
id	filename to retrieve the file
design_sequence	designed amino acids (may be subset of chain)
designed_chain_sequence	full sequence of the chain containing designed residues (recommended for synthesis)
num_design	number of designed residues
secondary_rank	intermediate rank from the sorting procedure
design_ptm	predicted TM score for intra-design contacts (higher = better)
design_iptm	predicted TM score for design–target contacts (higher = better)
design_to_target_iptm	same as design_iptm but for multi-chain designs
design_ipsae_min	min interaction pSAE: PAE-based confidence score for interaction (higher = better)
design_to_target_ipsae	interaction pSAE between design and target (higher = better)
min_design_to_target_pae	minimum PAE between design & target (lower = better)
plip_saltbridge	number of salt-bridge interactions
plip_hbonds	number of hydrogen-bond interactions
plip_hydrophobic	number of hydrophobic interactions
delta_sasa_original	ΔSASA when binder present vs absent
delta_sasa_refolded	same as above but on the refolded structure

Filtering & Sorting Criteria

This section first lists threshold filters that every design must pass (Filtering Criteria table)then explains how designs are ranked (Sorting Criteria table) weighted by their inverse importance.

Filtering Criteria

feature	lower_is_better	threshold	Pass
has_x	True	0.0	60
filter_rmsd	True	2.5	10
filter_rmsd_design	True	2.5	47
designfolding-filter_rmsd	True	2.5	49
ALA_fraction	True	0.3	60
GLY_fraction	True	0.3	60
GLU_fraction	True	0.3	60
LEU_fraction	True	0.3	60
VAL_fraction	True	0.3	60

Sorting Criteria

Metric	Inverse Importance
design_to_target_iptm	1
design_ptm	1
neg_min_design_to_target_pae	1
plip_hbonds_refolded	2
plip_saltbridge_refolded	2
delta_sasa_refolded	2

Results Summary

Aggregate statistics

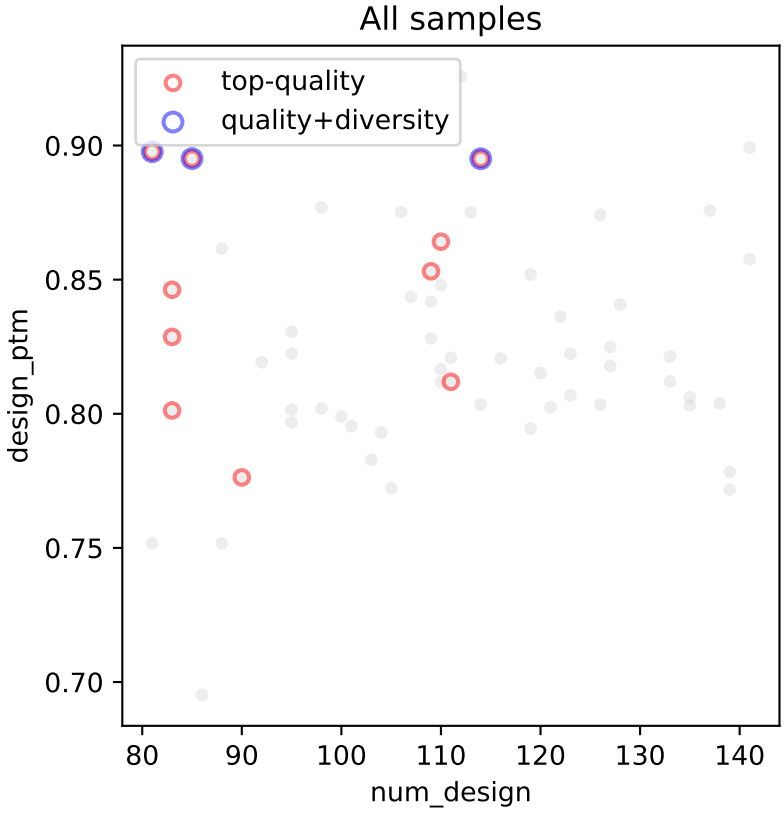
Quick numeric overview comparing (i) ALL incoming designs, (ii) the top-10 highest-quality designs, and (iii) the 3 quality+diversity designs produced by the lazy-greedy selection.

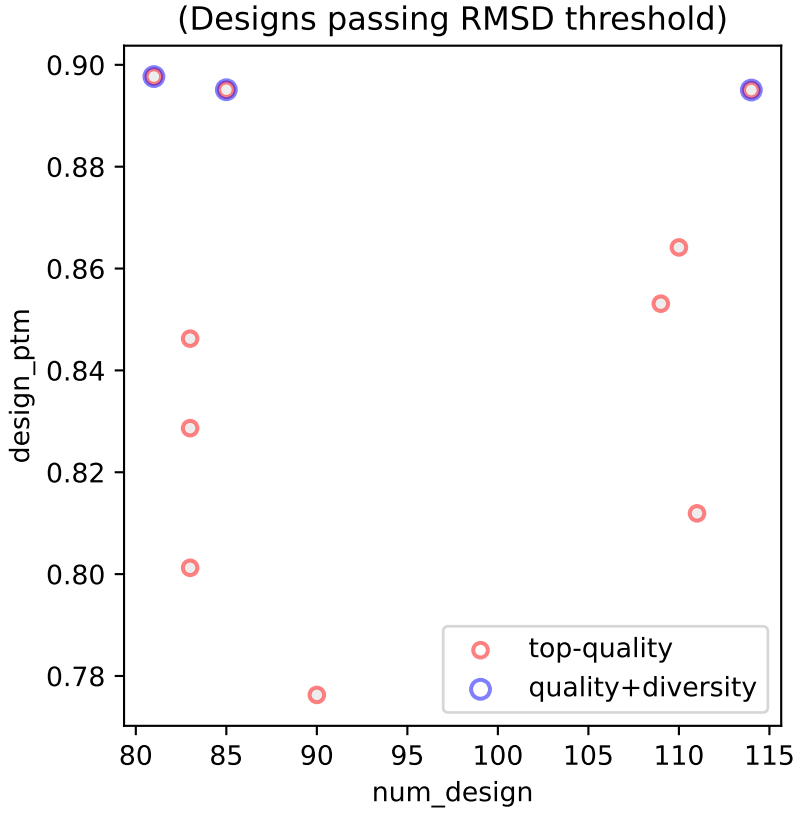
Metric	Mean	Mean top 10	Mean top 3 diverse
Num designs	60	-	-
design_ptm	0.824	0.847	0.896
neg_min_design_to_target_pae	-12.660	-9.689	-6.117
num_design	110.867	94.900	93.333
filter_rmsd	10.747	1.595	1.461
designfolding-filter_rmsd	1.858	0.918	0.974
design_iptm	0.325	0.412	0.543
design_to_target_iptm	0.325	0.412	0.543
min_design_to_target_pae	12.660	9.689	6.117
design_ipsae_min	0.033	0.055	0.119
design_to_target_ipsae	0.054	0.085	0.203
delta_sasa_refolded	817.183	839.851	907.682
plip_saltbridge_refolded	1.400	1.500	1.333
plip_hbonds_refolded	2.550	3.000	3.333

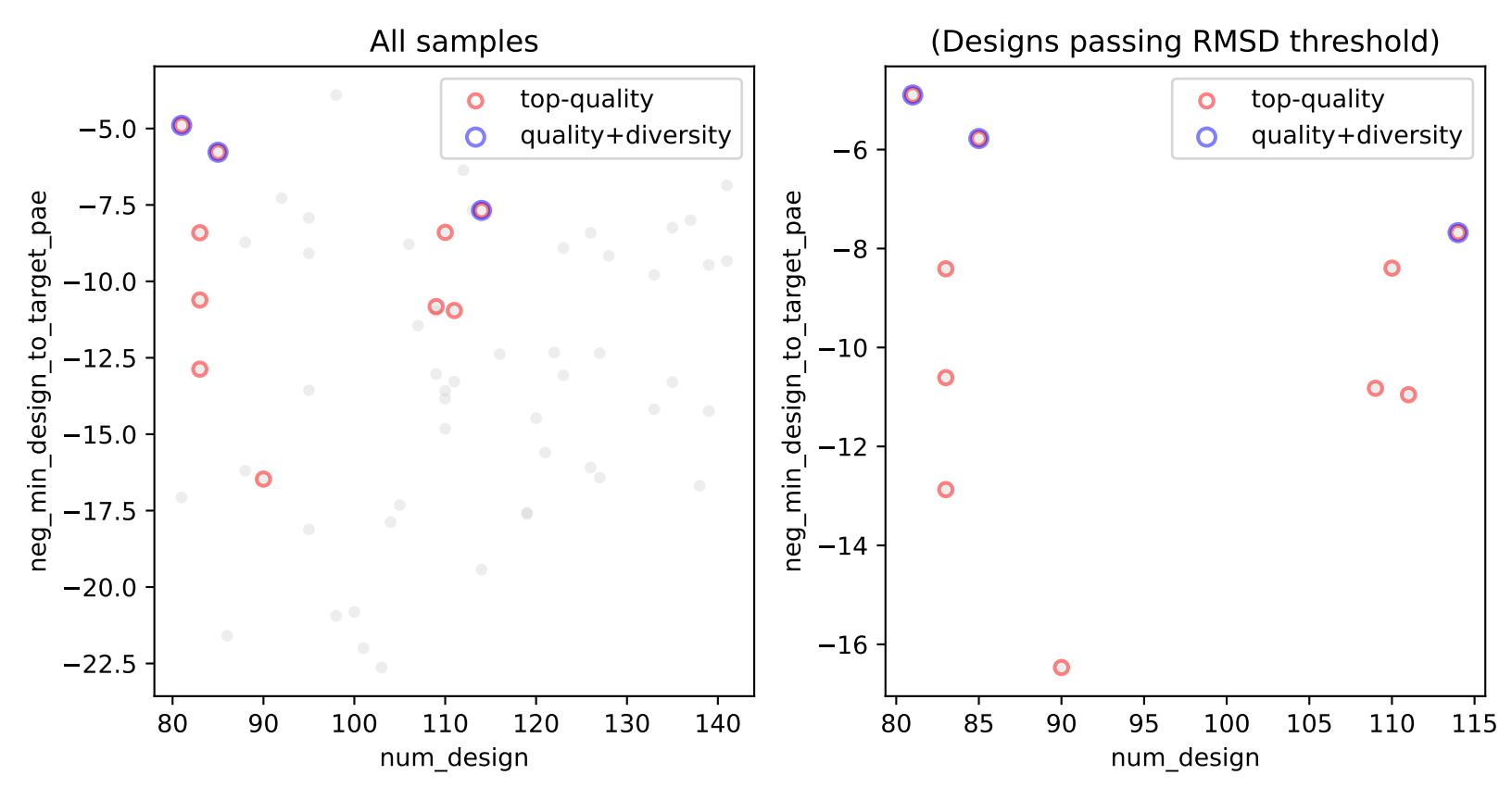
Scatter Plots

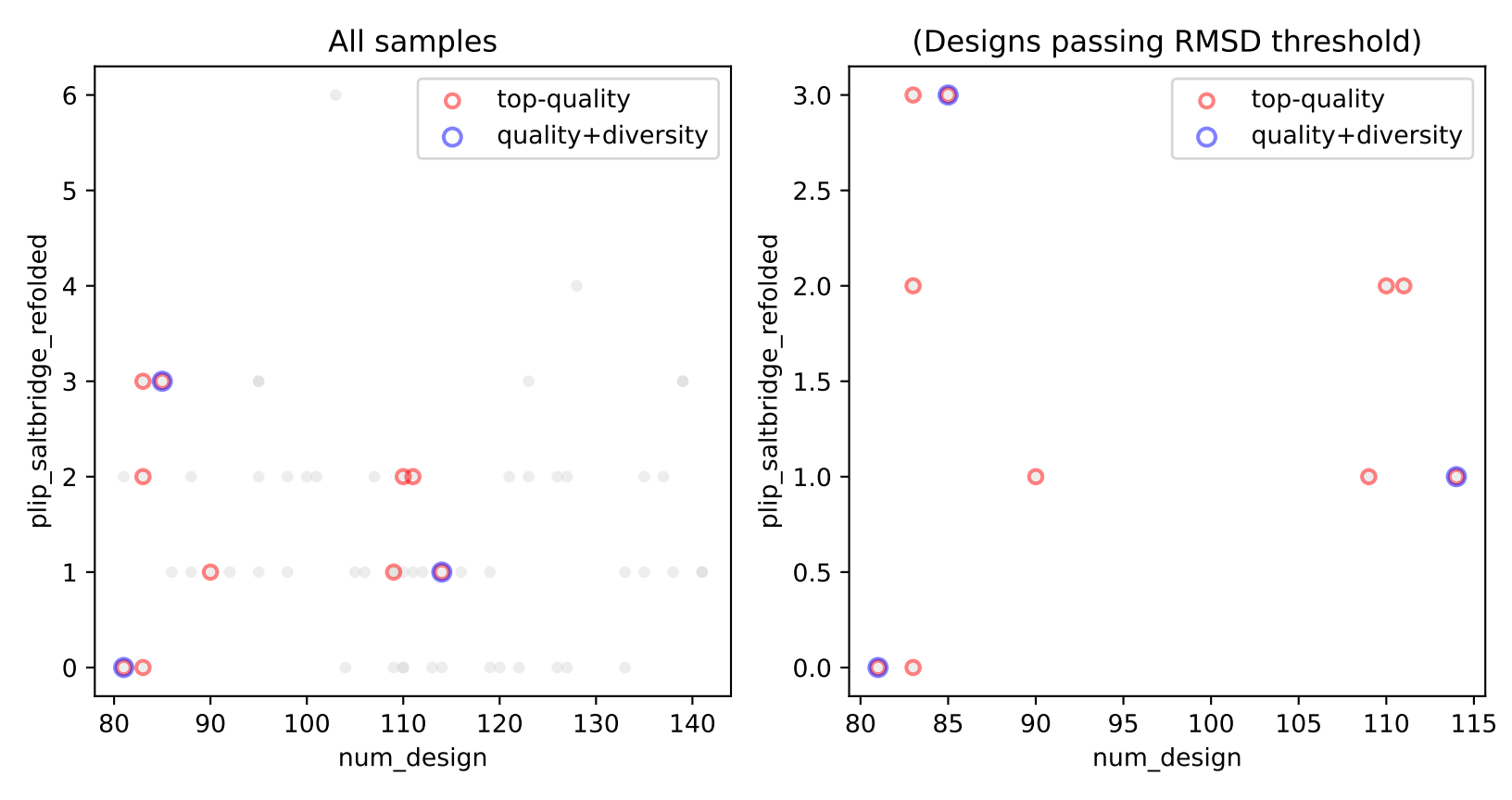
Metric relationships

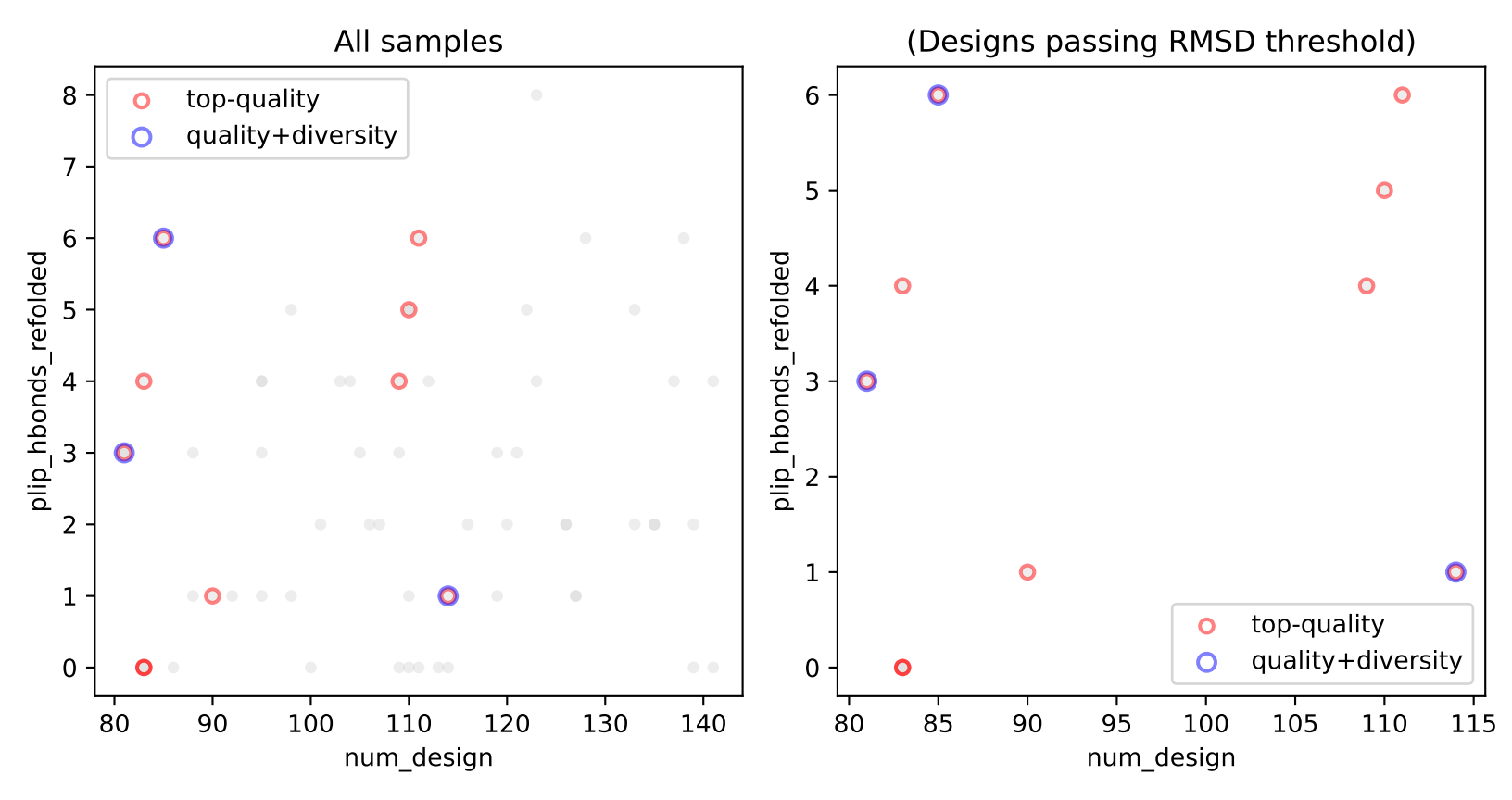
Each scatter page contains two panels:
• Left – all designs (grey) with overlays of Top (red) and Diverse (blue).
• Right – same but limited to designs passing the RMSD filter.

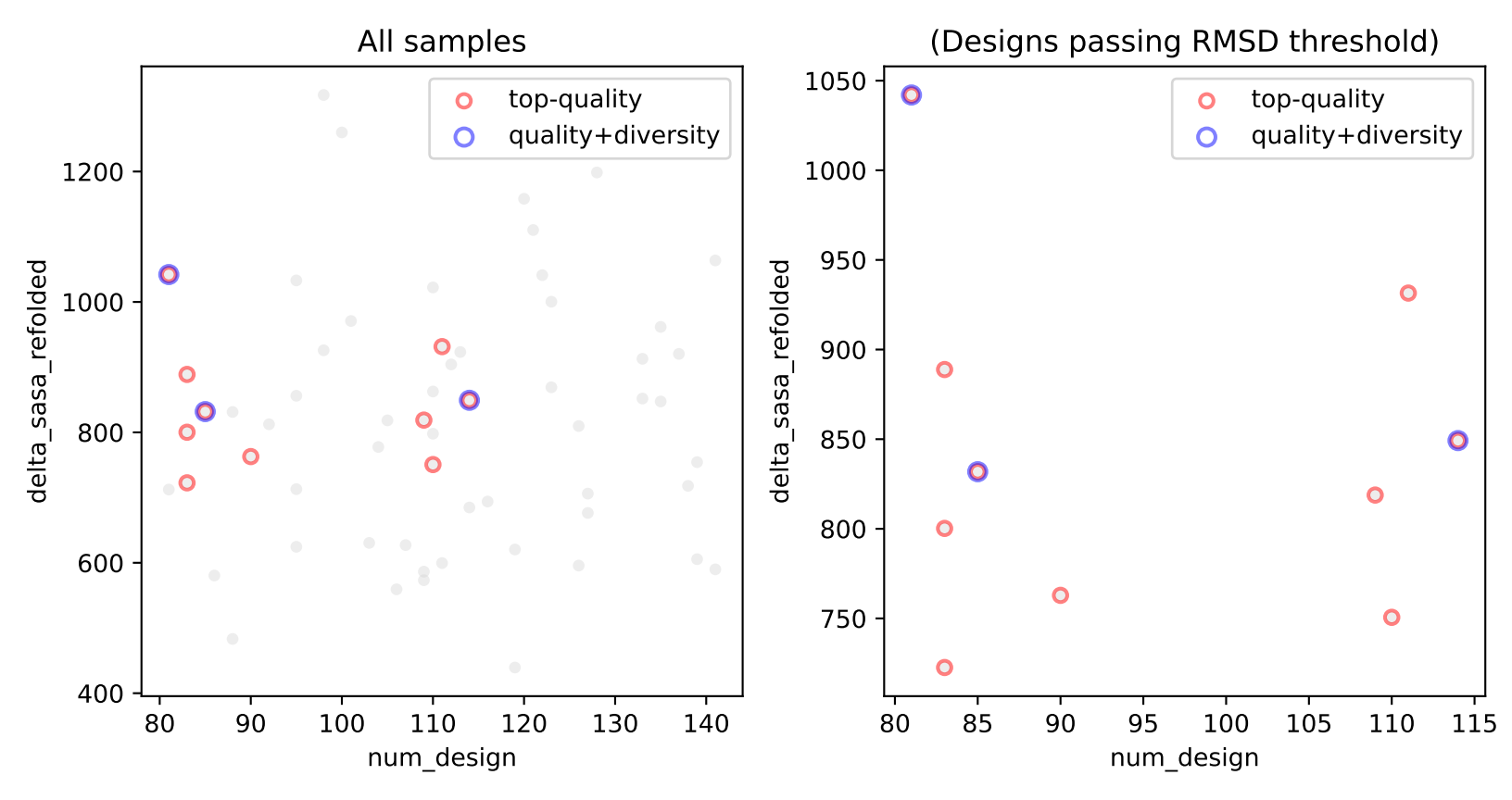

All samples
top-quality
quality+diversity
design_ptm
num_design
0.90
0.85
0.80
0.75
0.70
80
90
100
110
120
130
140

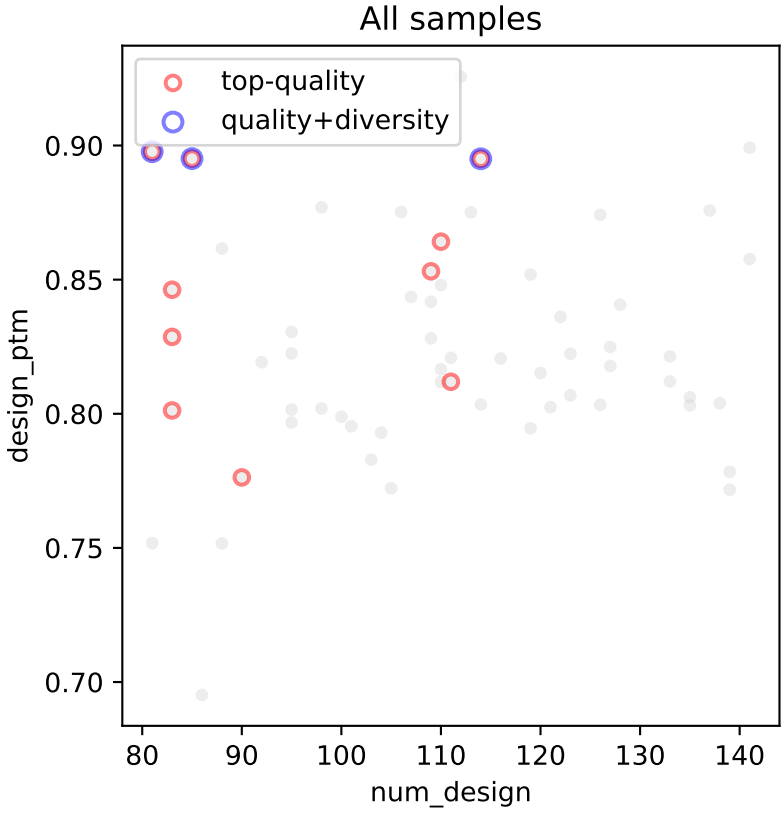

(Designs passing RMSD threshold)
top-quality
quality+diversity
design_ptm
num_design
0.90
0.88
0.86
0.84
0.82
0.80
0.78
80
85
90
95
100
105
110
115

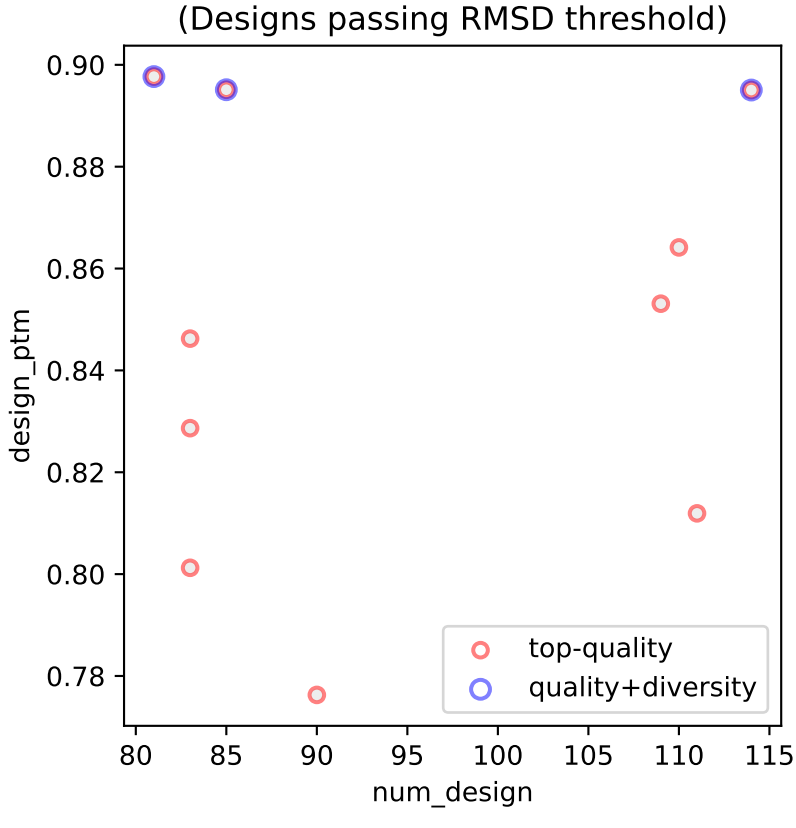

All samples
top-quality
quality+diversity
neg_min_design_to_target_pae
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
−22.5
80
90
100
110
120
130
140
num_design
(Designs passing RMSD threshold)
top-quality
quality+diversity
neg_min_design_to_target_pae
−6
−8
−10
−12
−14
−16
80
85
90
95
100
105
110
115
num_design

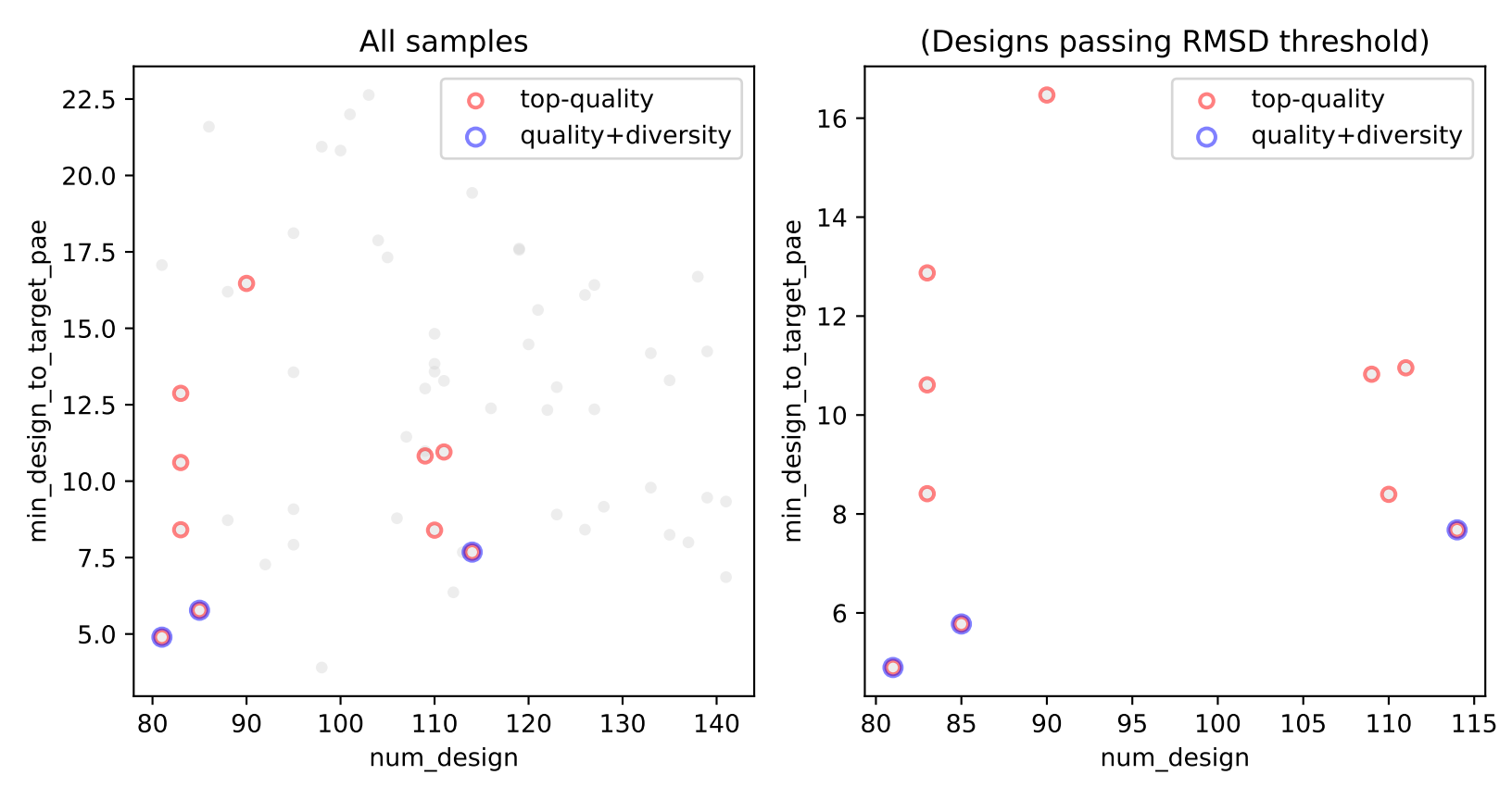

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
plip_saltbridge_refolded
num_design
0
1
2
3
4
5
6
0.0
0.5
1.0
1.5
2.0
2.5
3.0
80
85
90
95
100
105
110
115
120
130
140

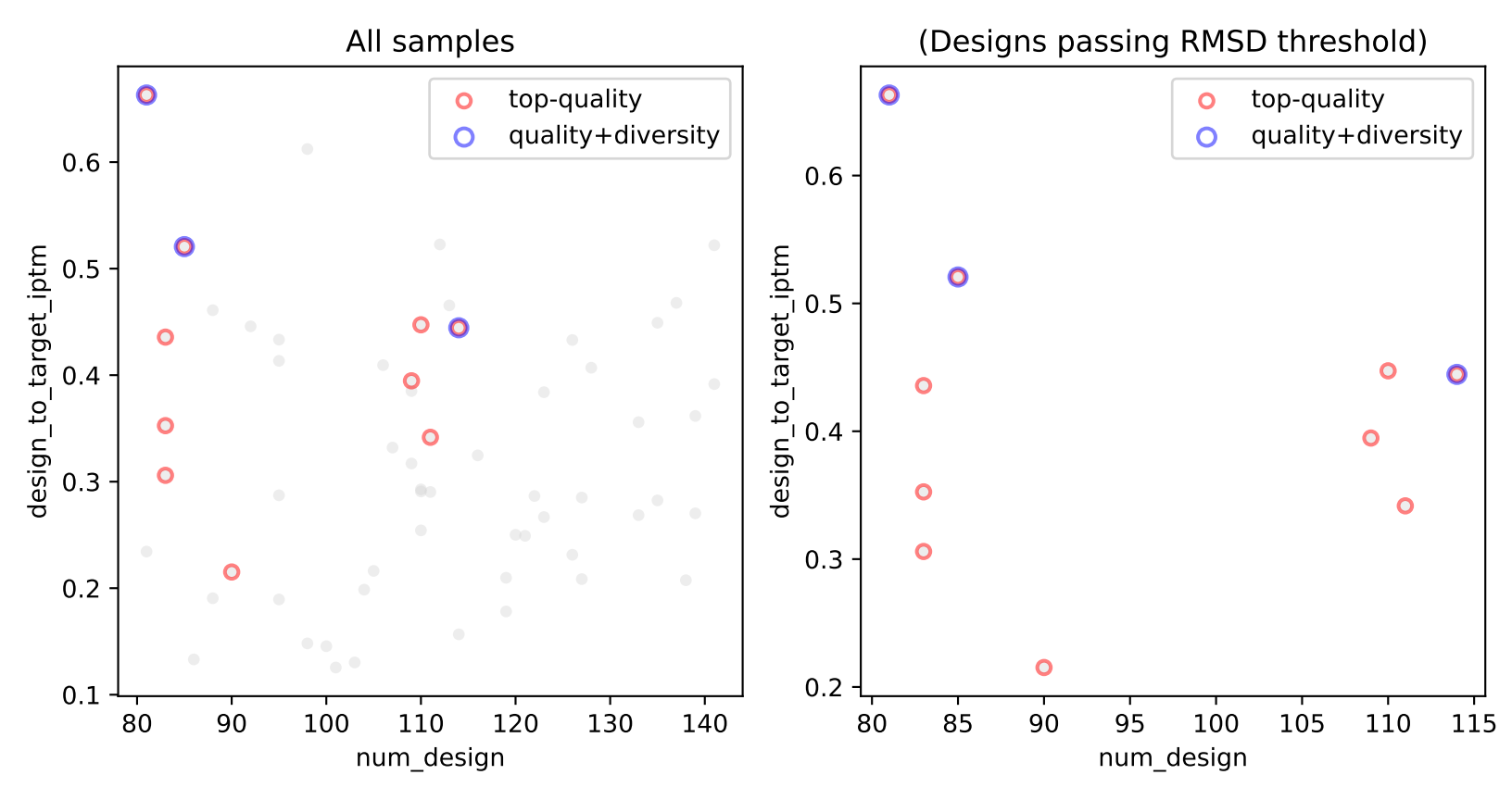

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
plip_hbonds_refolded
num_design
0
1
2
3
4
5
6
7
8
80
85
90
95
100
105
110
115
120
130
140

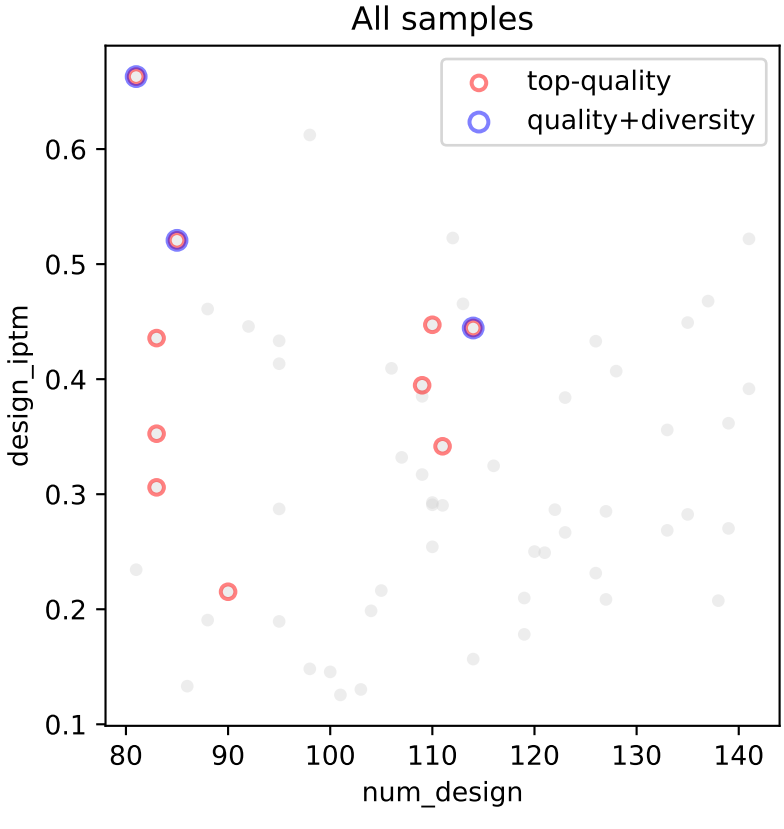

All samples
top-quality
quality+diversity
delta_sasa_refolded
num_design
1200
1000
800
600
400
80
90
100
110
120
130
140
(Designs passing RMSD threshold)
top-quality
quality+diversity
delta_sasa_refolded
num_design
1050
1000
950
900
850
800
750
80
85
90
95
100
105
110
115

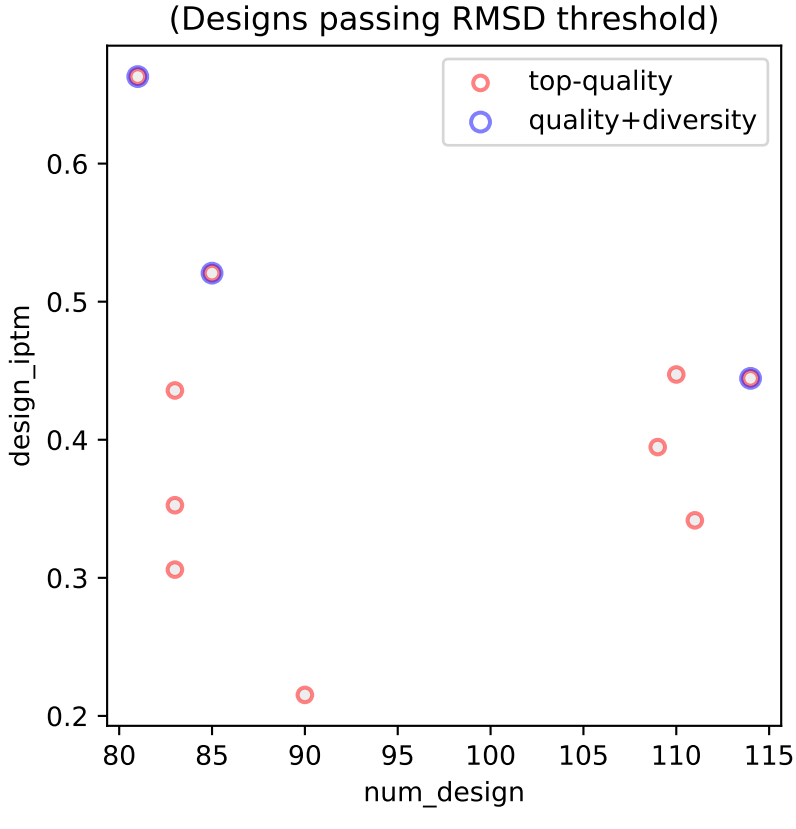

All samples
top-quality
quality+diversity
design_ptm
num_design
0.90
0.85
0.80
0.75
0.70
80
90
100
110
120
130
140

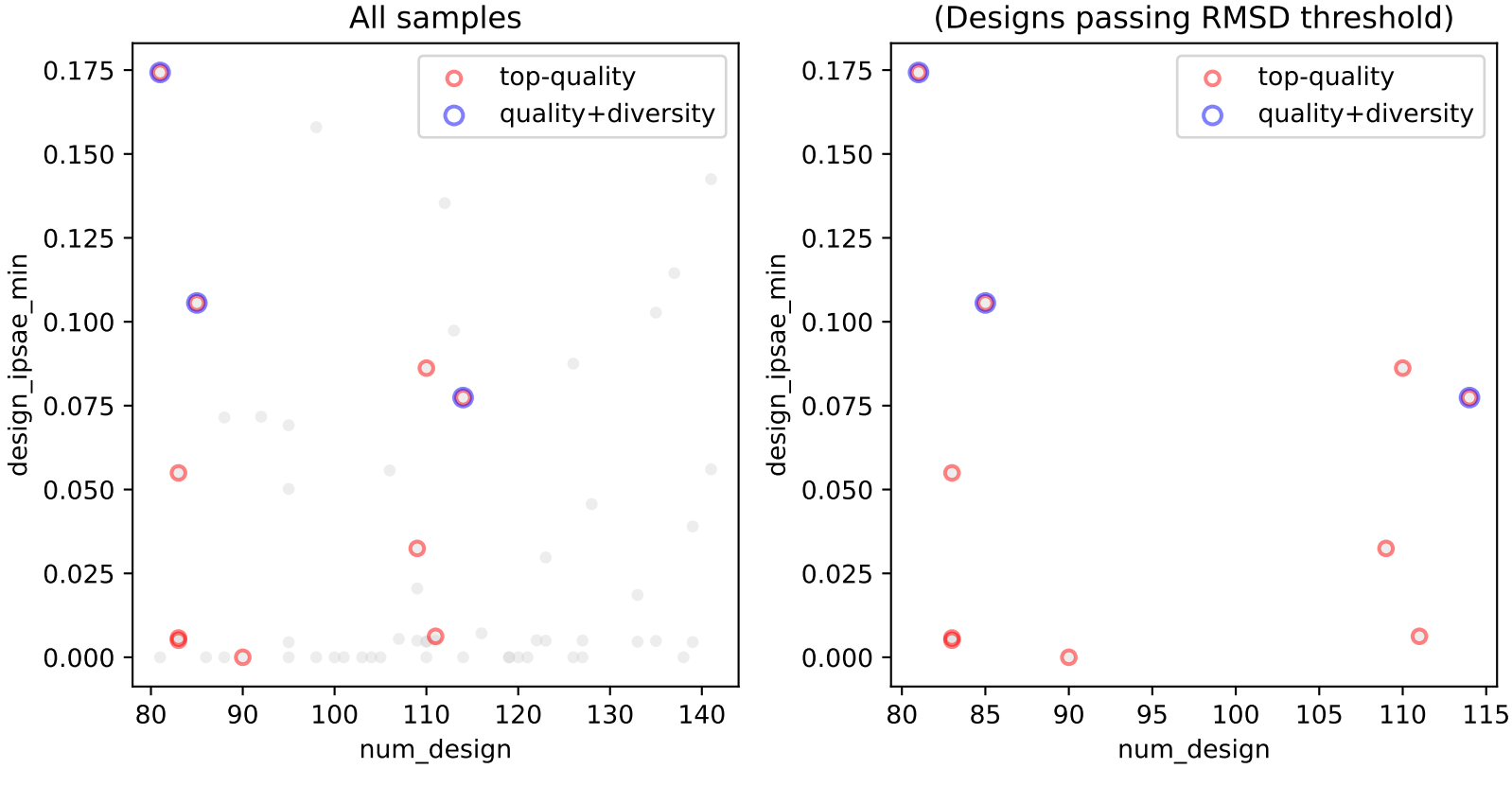

(Designs passing RMSD threshold)
top-quality
quality+diversity
design_ptm
num_design
0.90
0.88
0.86
0.84
0.82
0.80
0.78
80
85
90
95
100
105
110
115

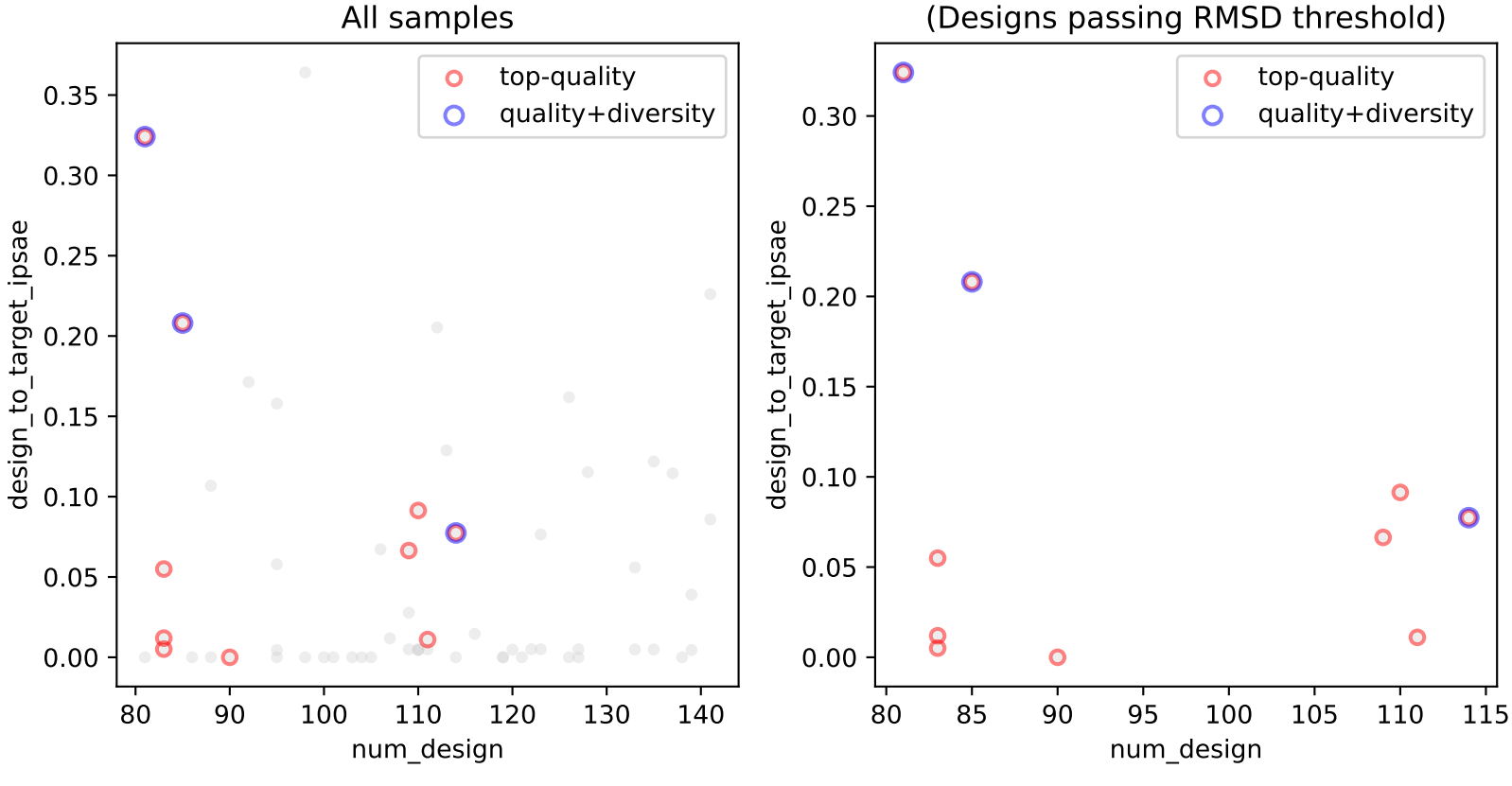

All samples
top-quality
quality+diversity
min_design_to_target_pae
num_design
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
80
90
100
110
120
130
140
(Designs passing RMSD threshold)
top-quality
quality+diversity
min_design_to_target_pae
num_design
16
14
12
10
8
6
80
85
90
95
100
105
110
115

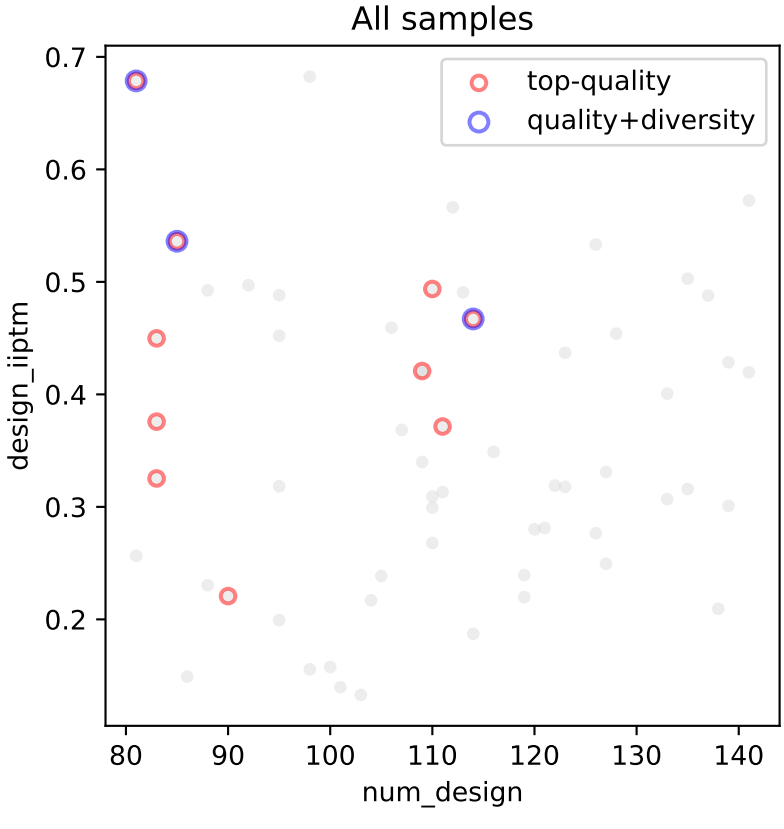

All samples
top-quality
quality+diversity
design_to_target_iptm
num_design
0.1
0.2
0.3
0.4
0.5
0.6
80
90
100
110
120
130
140
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_to_target_iptm
num_design
0.2
0.3
0.4
0.5
0.6
80
85
90
95
100
105
110
115

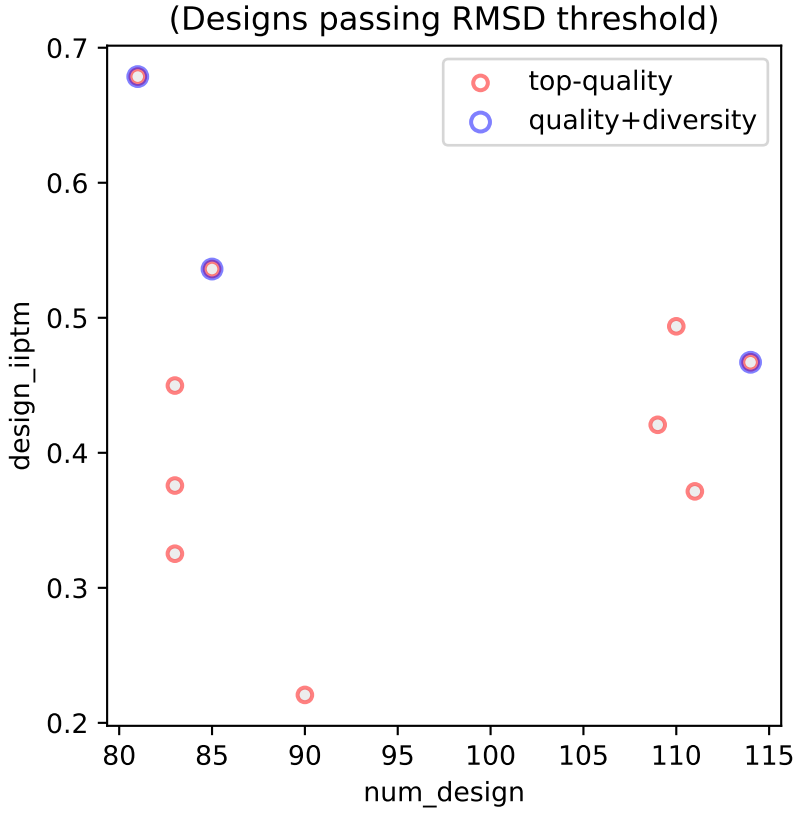

All samples
top-quality
quality+diversity
design_iptm
num_design
0.1
0.2
0.3
0.4
0.5
0.6
80
90
100
110
120
130
140

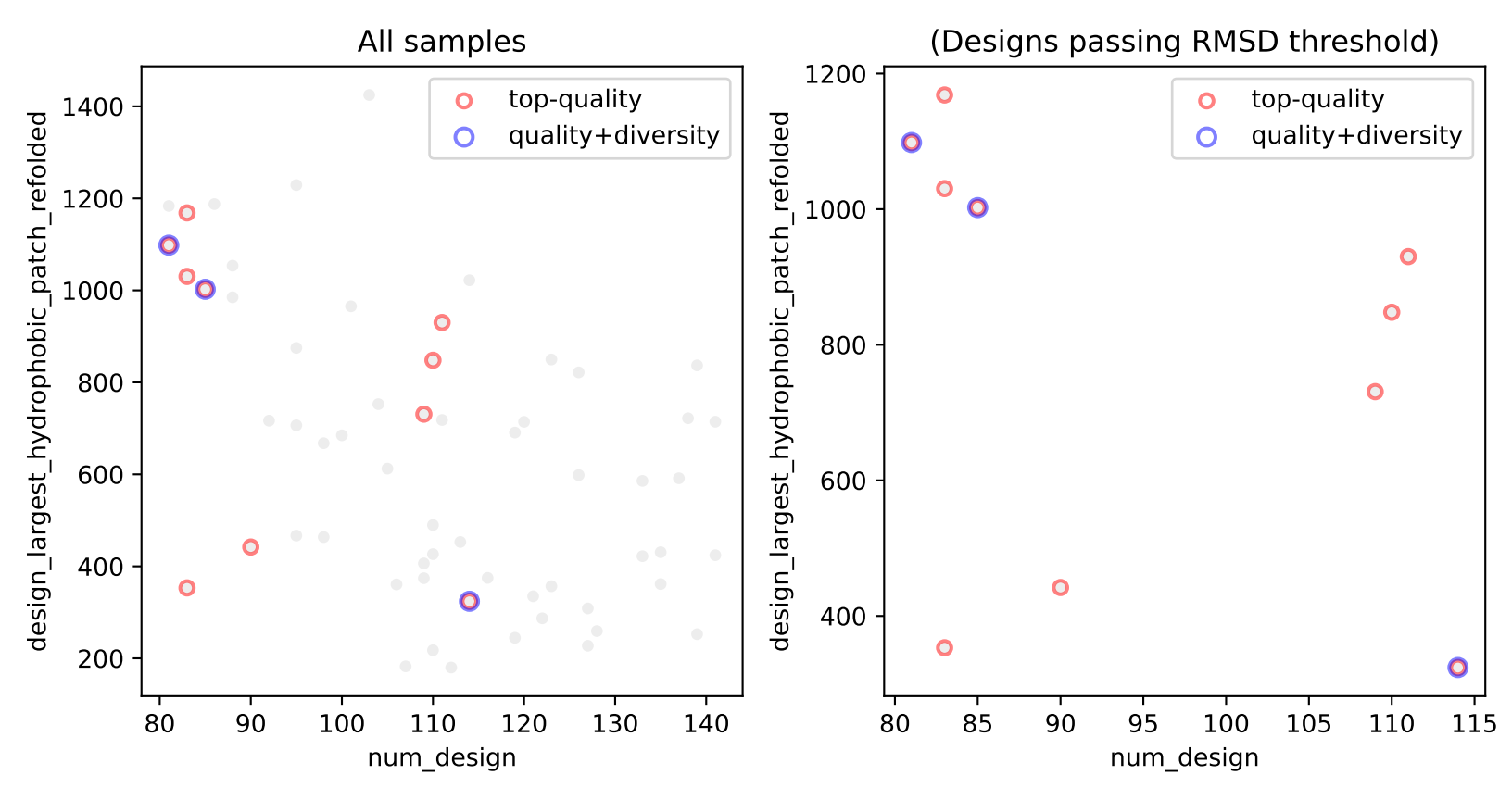

(Designs passing RMSD threshold)
top-quality
quality+diversity
design_iptm
num_design
0.2
0.3
0.4
0.5
0.6
80
85
90
95
100
105
110
115

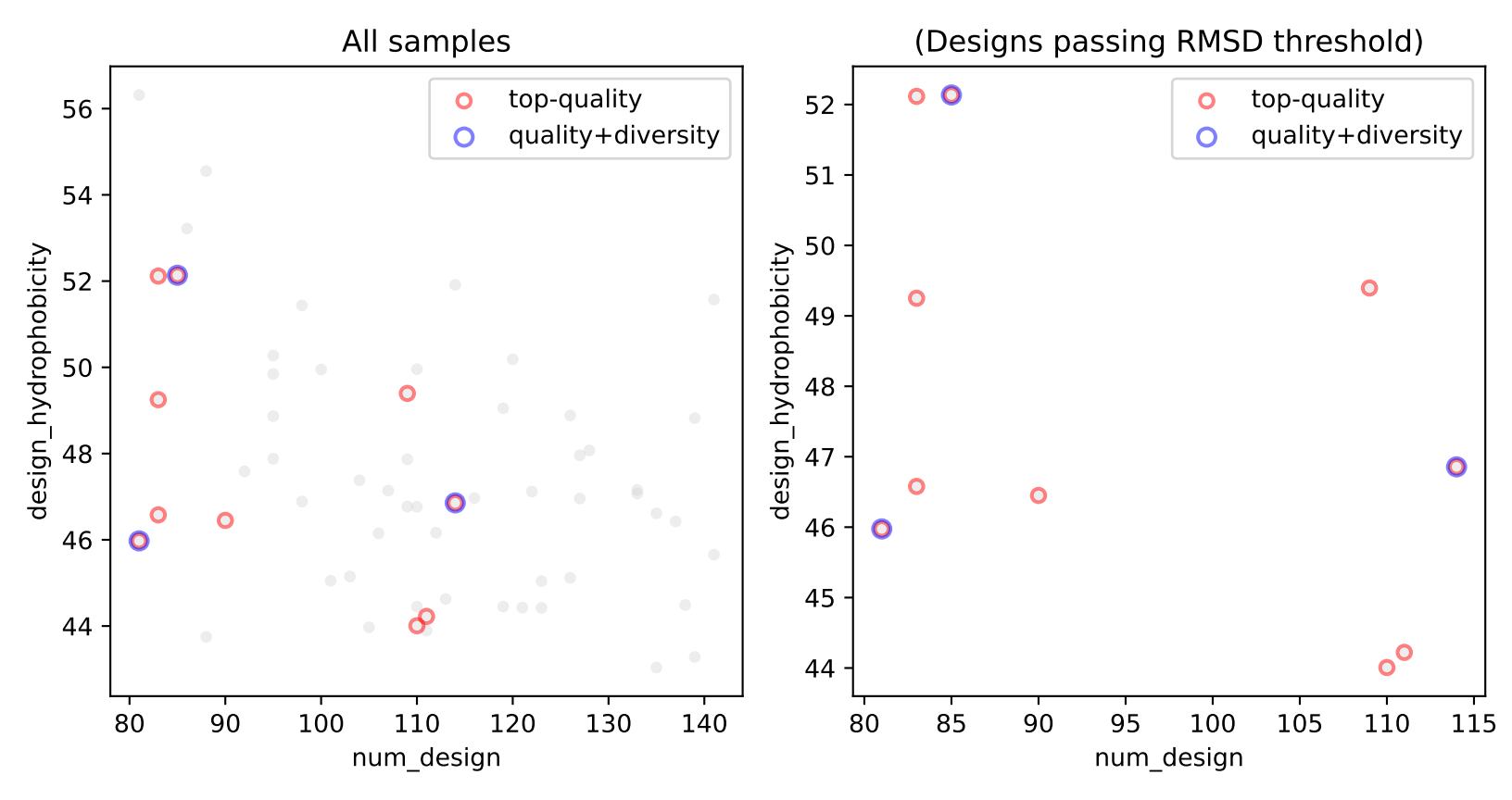

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_ipsae_min
num_design
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
80
90
100
110
120
130
140
85
95
105
115


All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_to_target_ipsae
num_design
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
80
85
90
95
100
105
110
115
120
130
140


All samples
top-quality
quality+diversity
design_iiptm
num_design
0.7
0.6
0.5
0.4
0.3
0.2
80
90
100
110
120
130
140


(Designs passing RMSD threshold)
top-quality
quality+diversity
design_iiptm
num_design
0.7
0.6
0.5
0.4
0.3
0.2
80
85
90
95
100
105
110
115

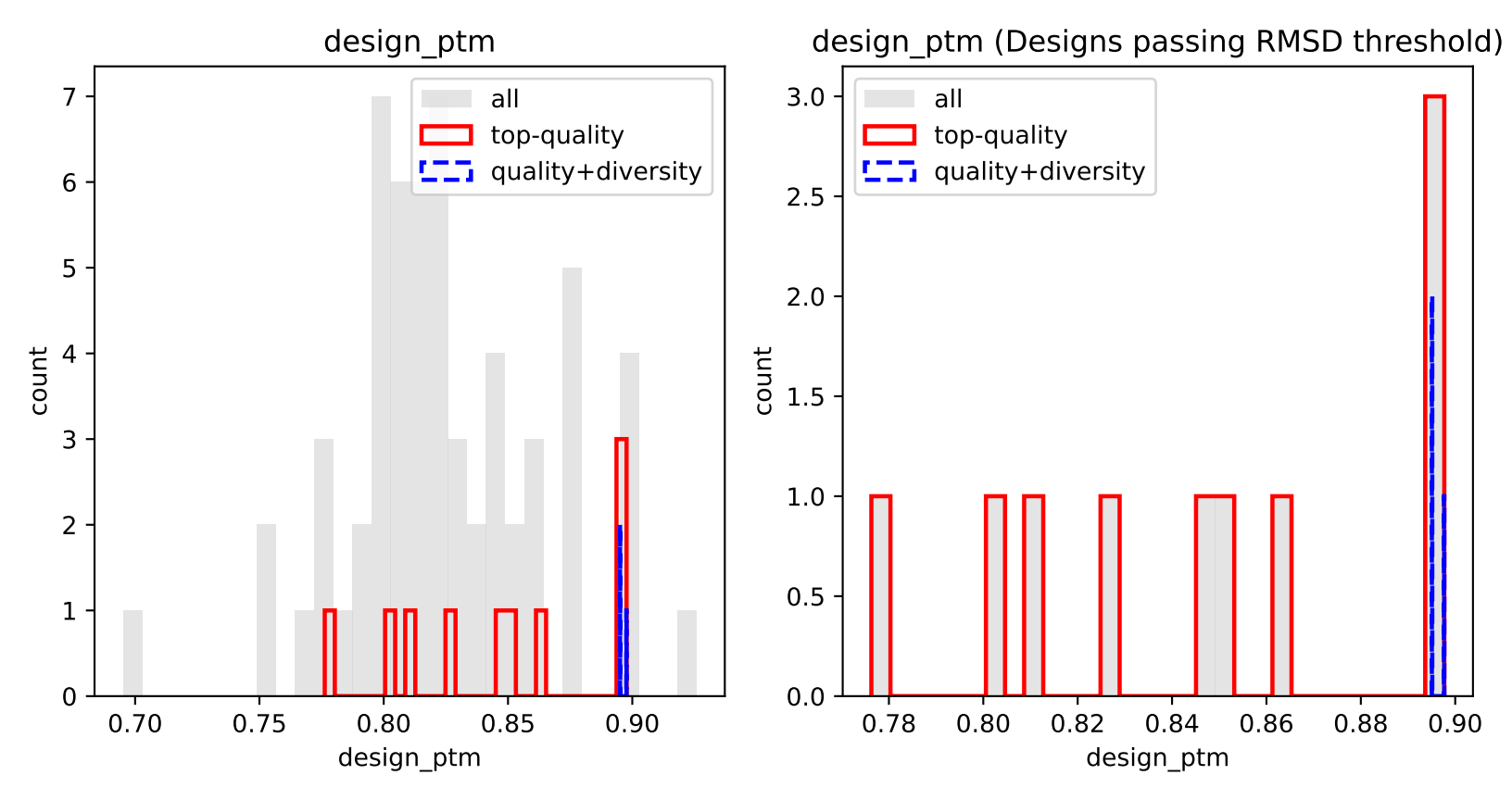

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_largest_hydrophobic_patch_refolded
num_design
1400
1200
1000
800
600
400
200
80
90
100
110
120
130
140
85
95
105
115

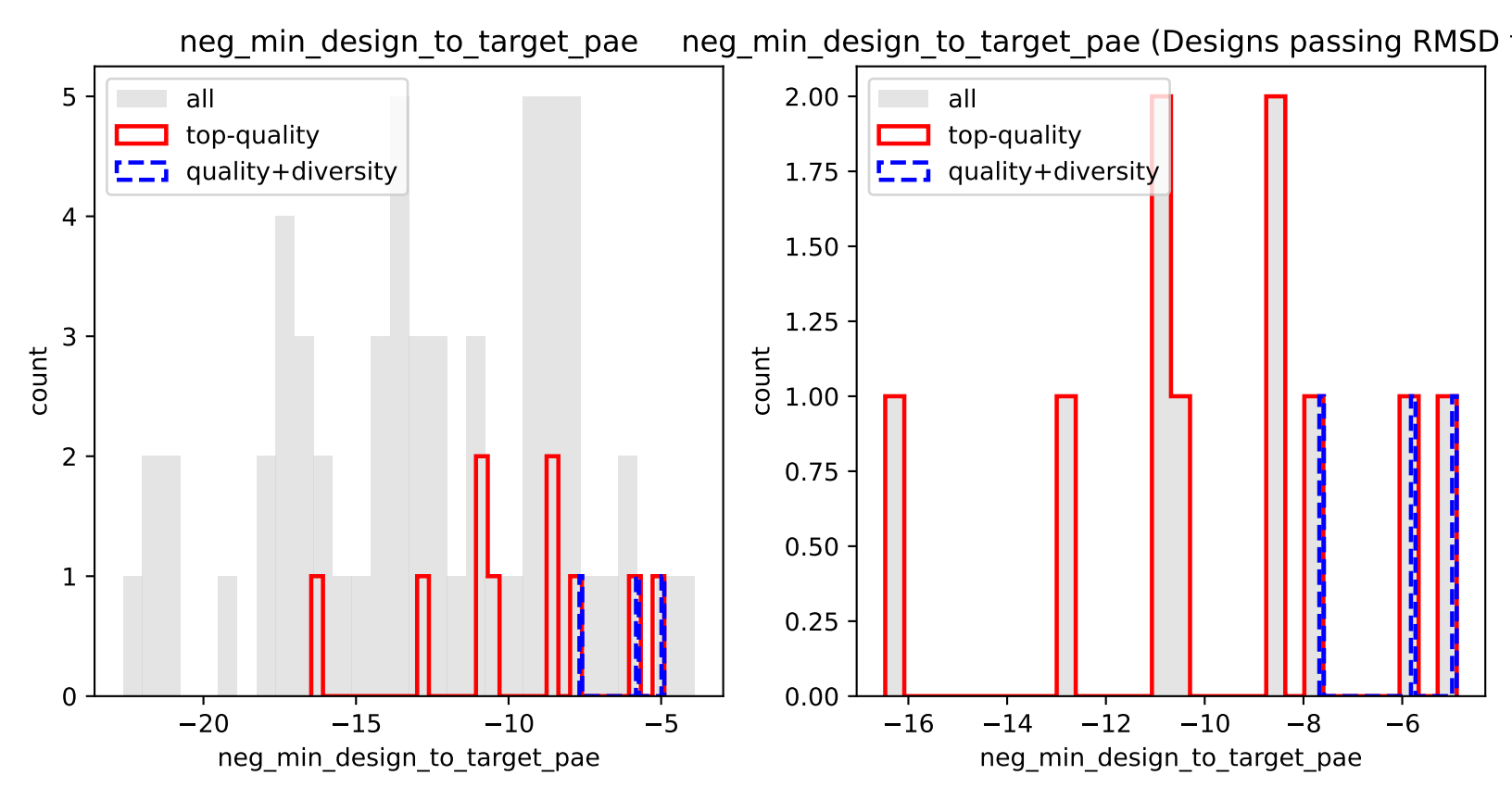

All samples
top-quality
quality+diversity
design_hydrophobicity
num_design
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_hydrophobicity
num_design

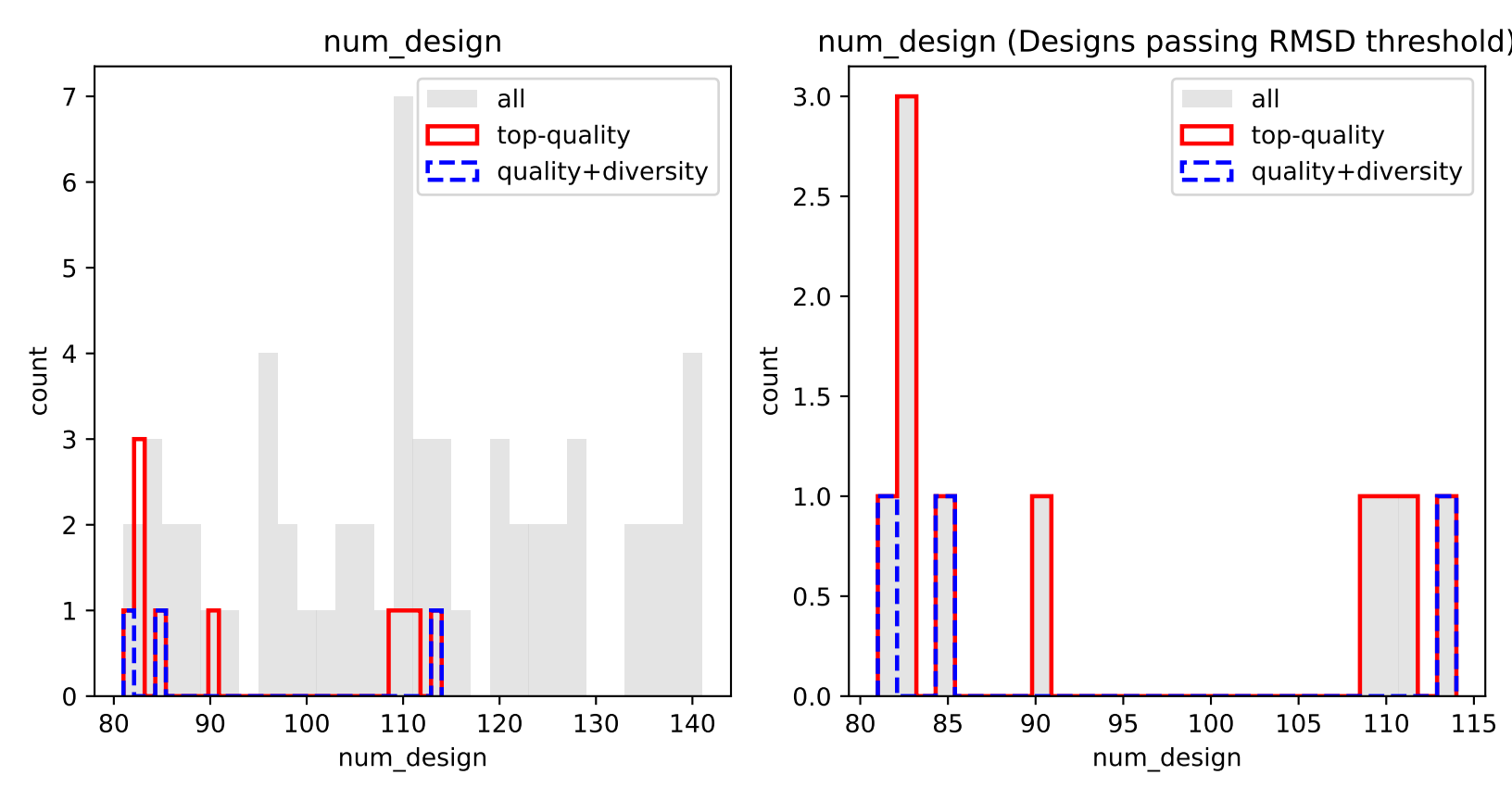
Metric Distributions

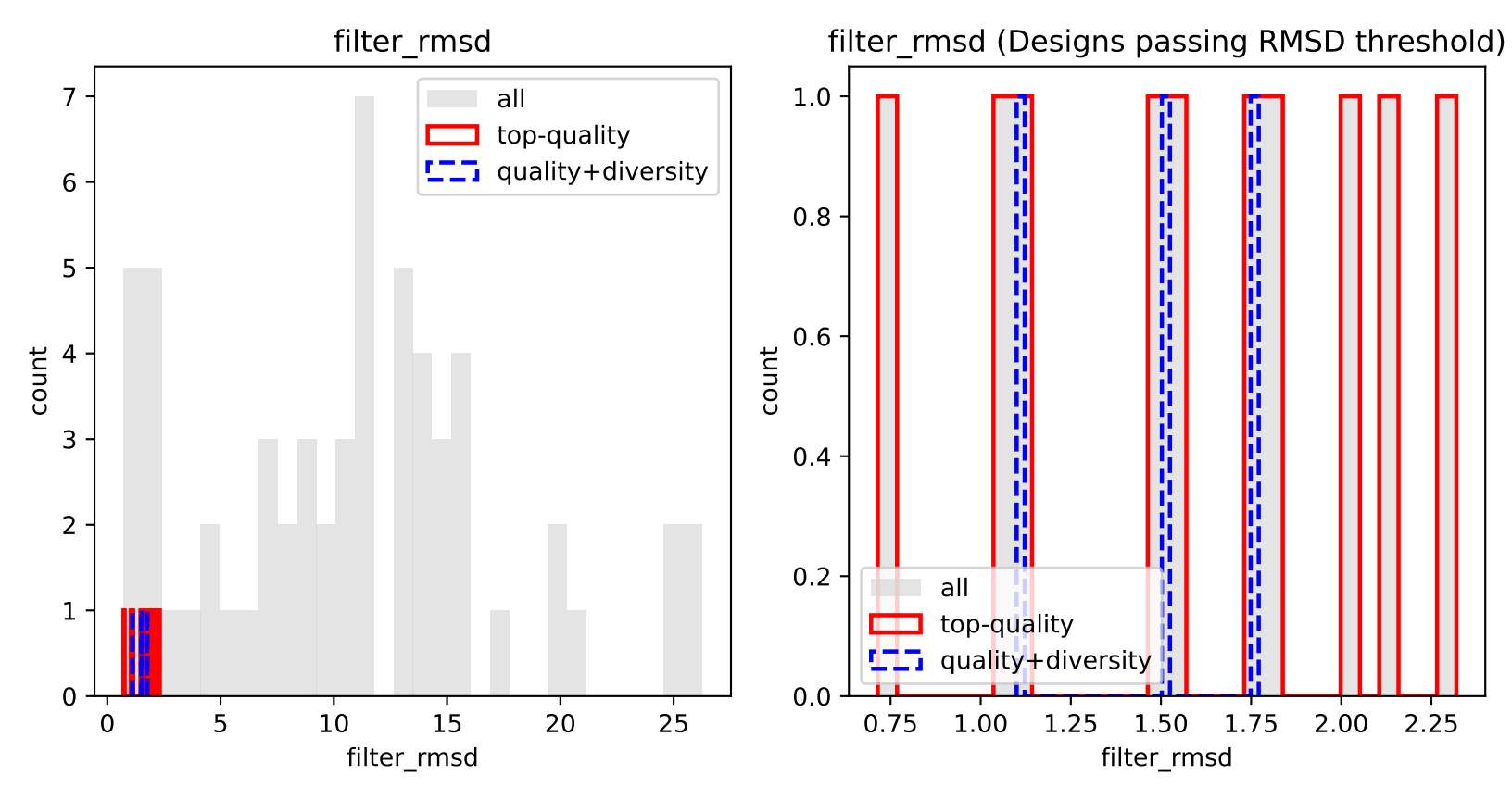
Histograms

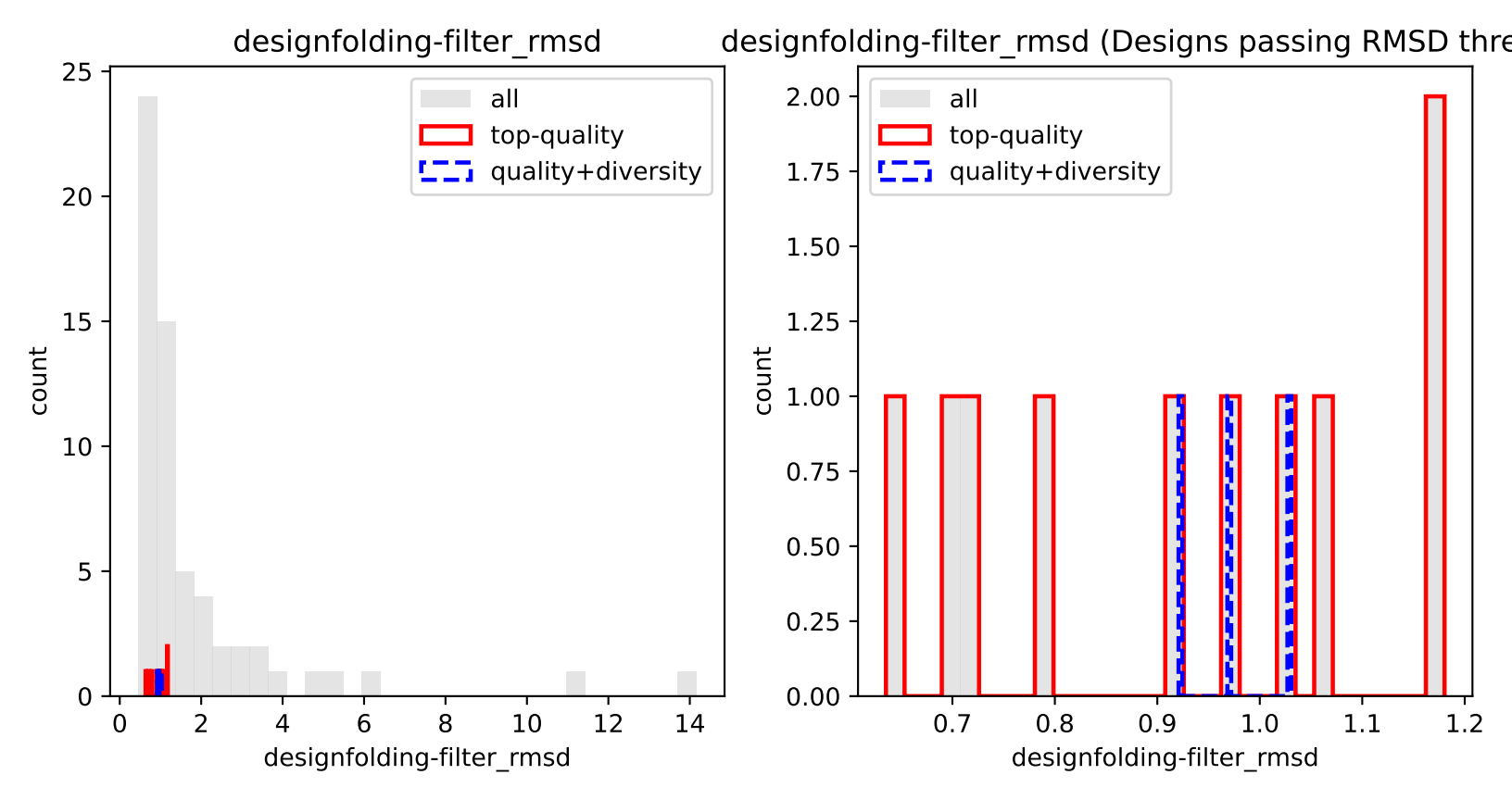
Distribution of each metric across all designs (grey) with overlays for Top-quality (red outline) and Diversity-optimised (blue dashed) subsets. The right panel repeats the histogram but only for designs that pass the RMSD threshold.

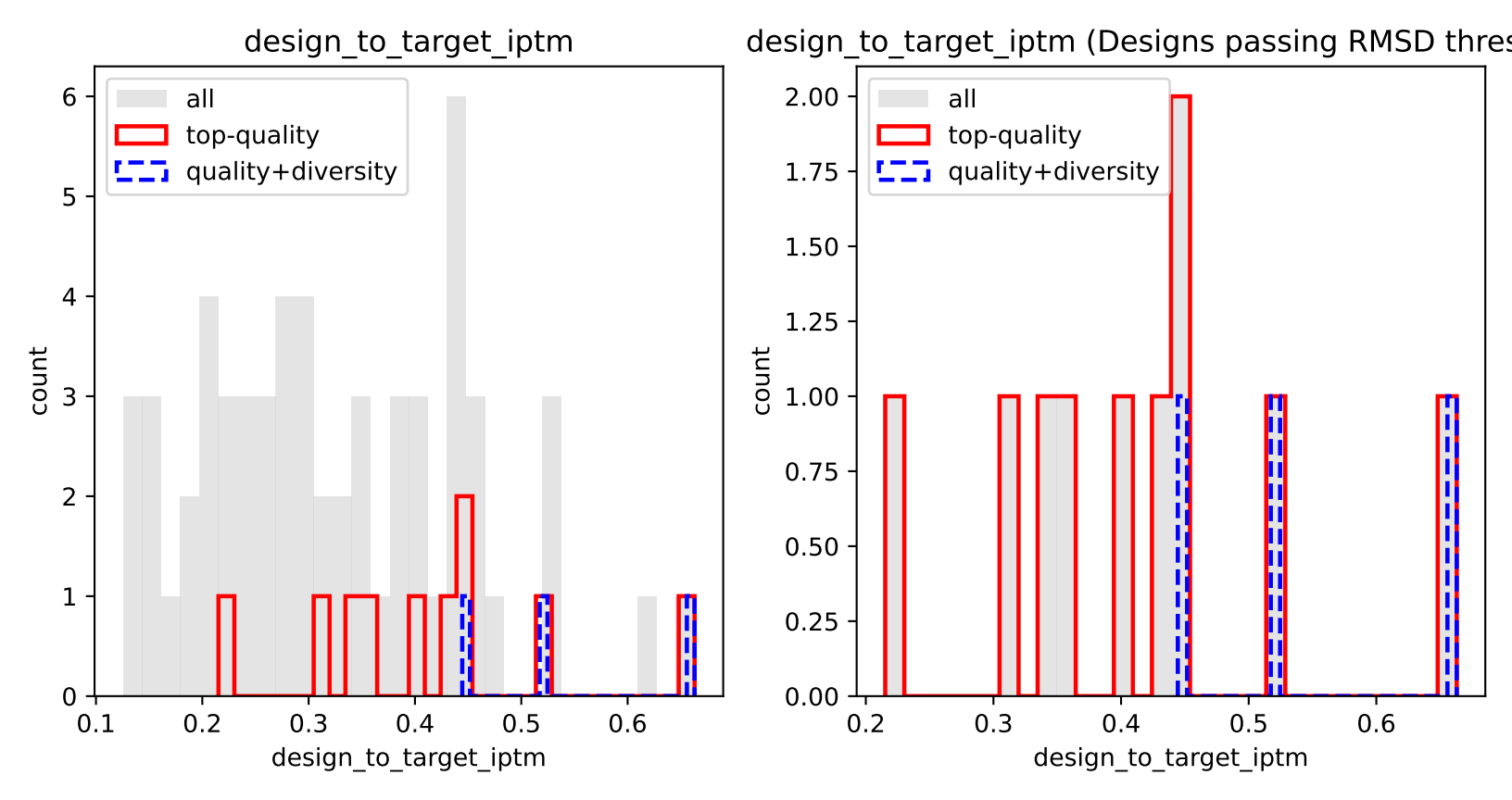

design_ptm
all
top-quality
quality+diversity
count
design_ptm
0.70
0.75
0.80
0.85
0.90
0
1
2
3
4
5
6
7
design_ptm (Designs passing RMSD threshold)
all
top-quality
quality+diversity
count
design_ptm
0.78
0.80
0.82
0.84
0.86
0.88
0.90
0.0
0.5
1.0
1.5
2.0
2.5
3.0

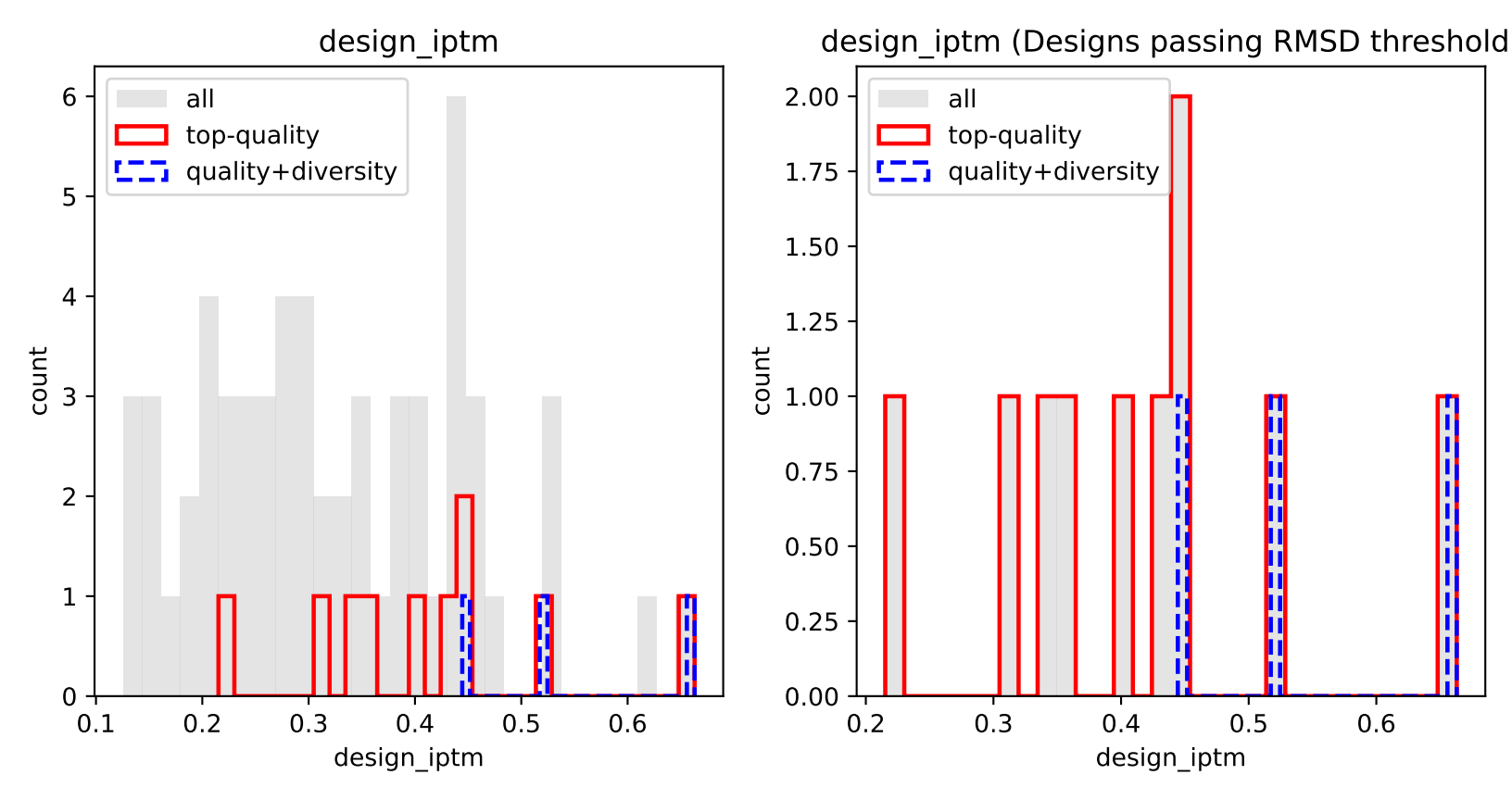

neg_min_design_to_target_pae
neg_min_design_to_target_pae (Designs passing RMSD
all
top-quality
quality+diversity
count
neg_min_design_to_target_pae
0
1
2
3
4
5
−20
−15
−10
−5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
−16
−14
−12
−10
−8
−6

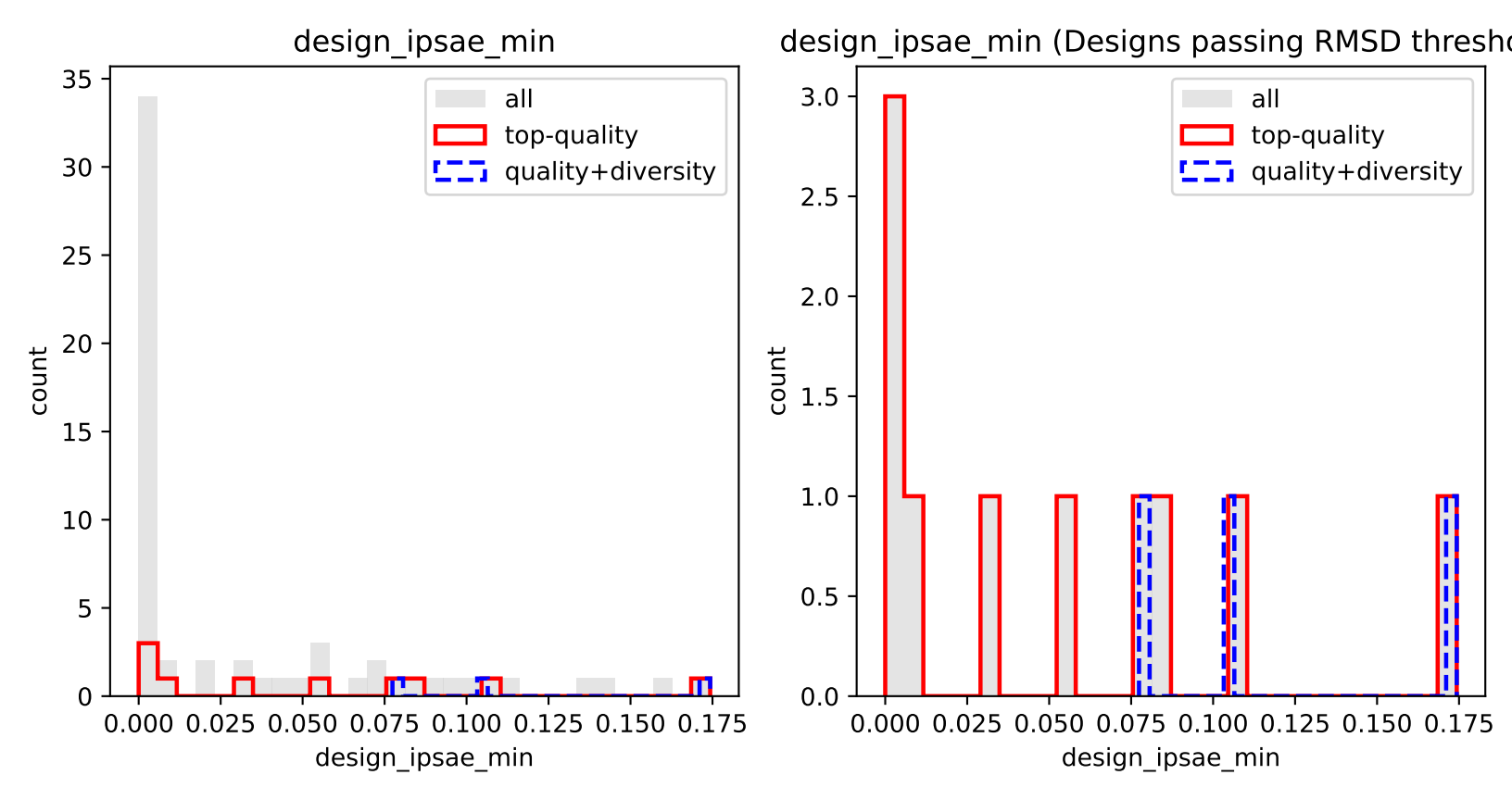

num_design
all
top-quality
quality+diversity
count
num_design
num_design (Designs passing RMSD threshold)
all
top-quality
quality+diversity
count
num_design

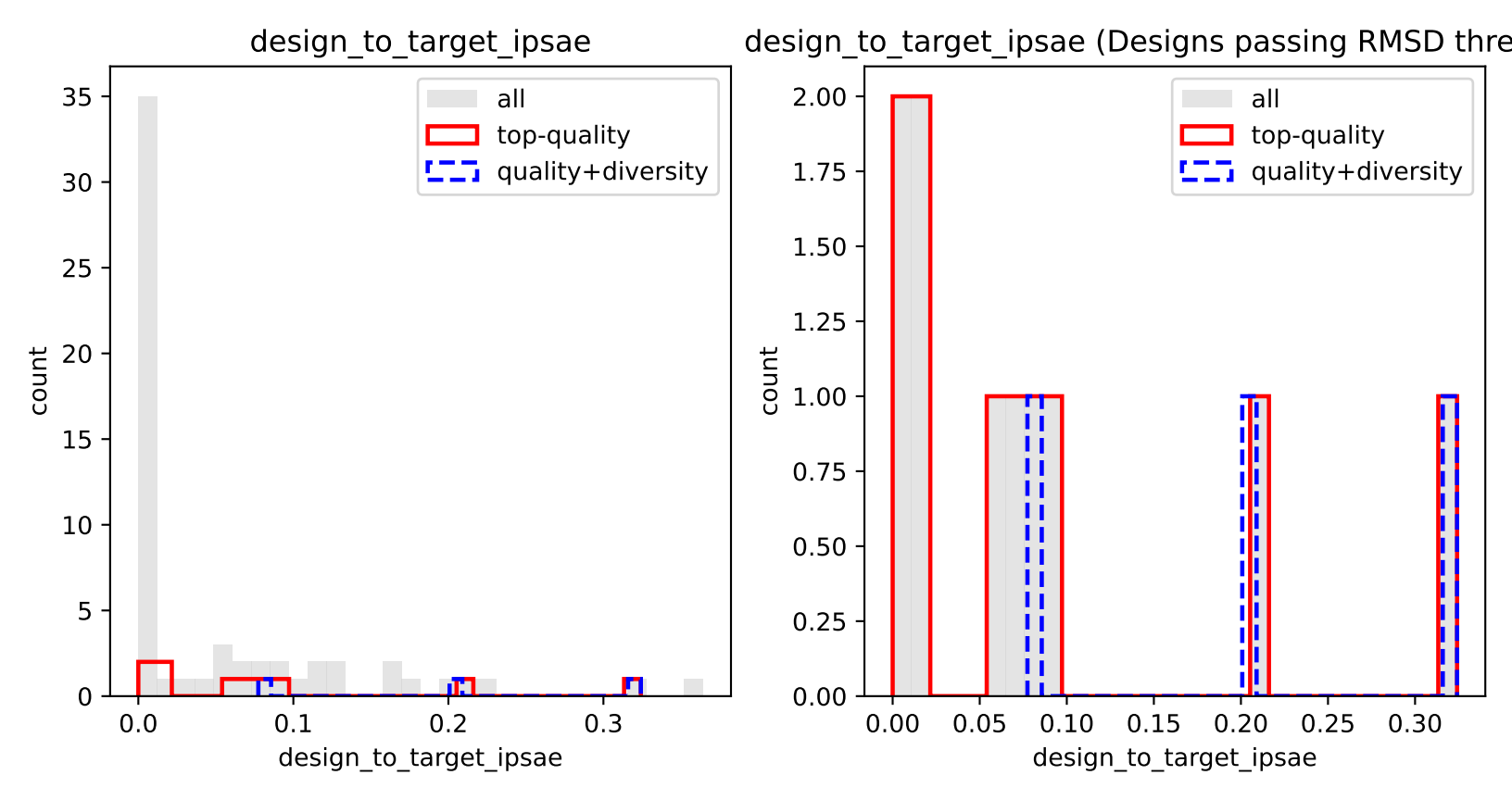

filter_rmsd
all
top-quality
quality+diversity
count
filter_rmsd
filter_rmsd (Designs passing RMSD threshold)
all
top-quality
quality+diversity
count
filter_rmsd

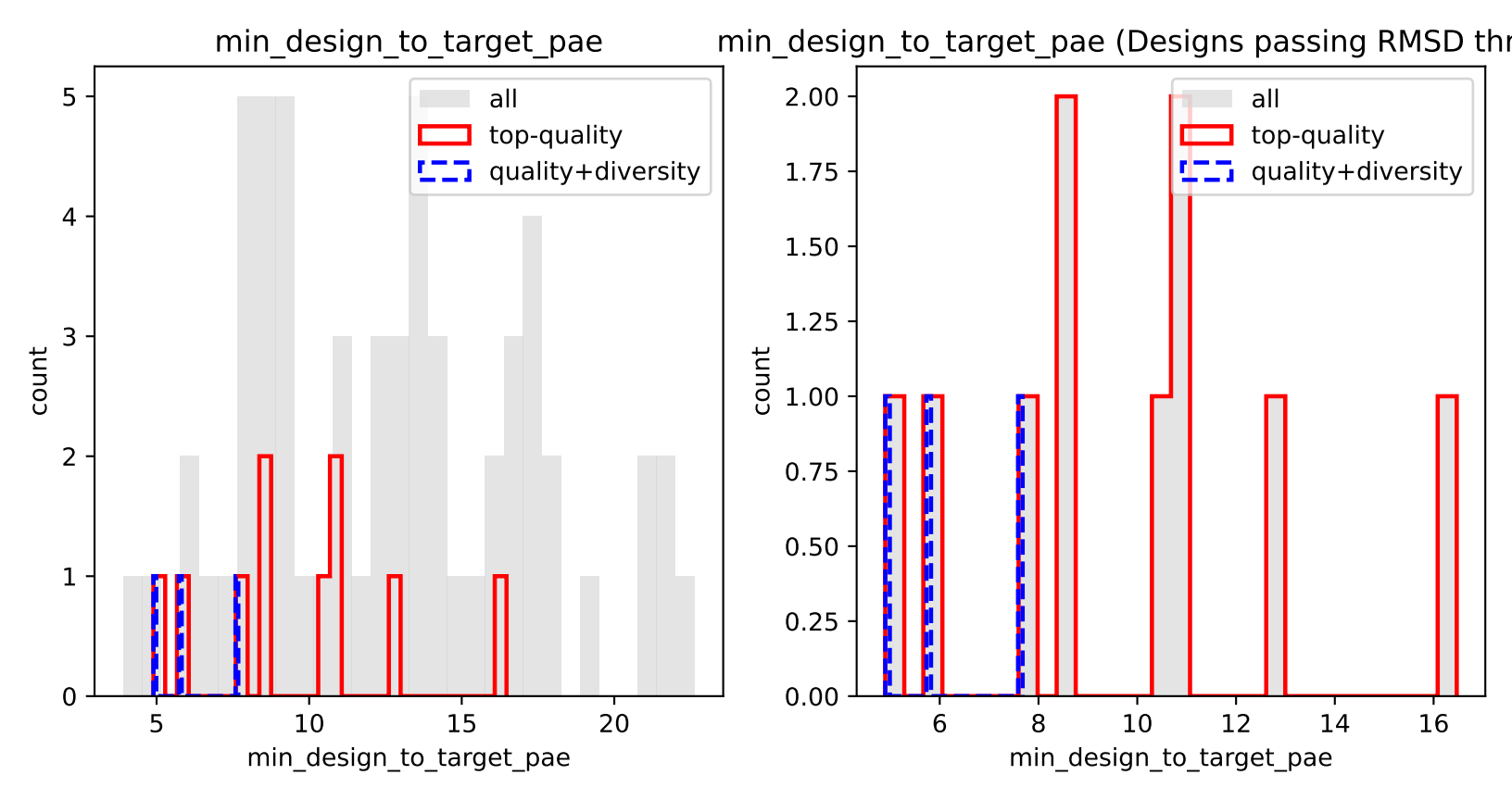

designfolding-filter_rmsd
all
top-quality
quality+diversity
count
designfolding-filter_rmsd
designfolding-filter_rmsd (Designs passing RMSD thre
all
top-quality
quality+diversity
count
designfolding-filter_rmsd

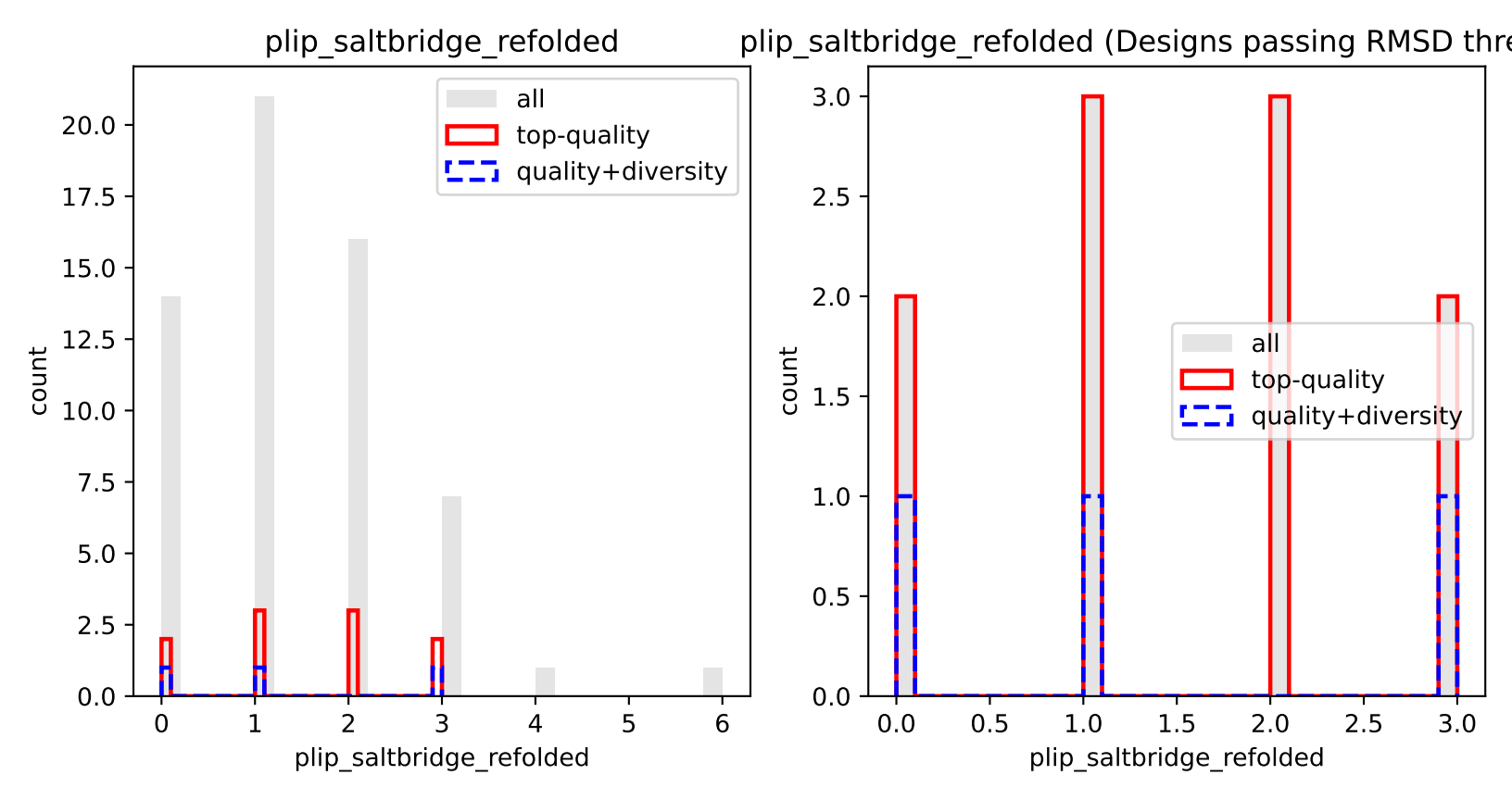

design_to_target_iptm
all
top-quality
quality+diversity
count
design_to_target_iptm
design_to_target_iptm (Designs passing RMSD thres
all
top-quality
quality+diversity
count
design_to_target_iptm

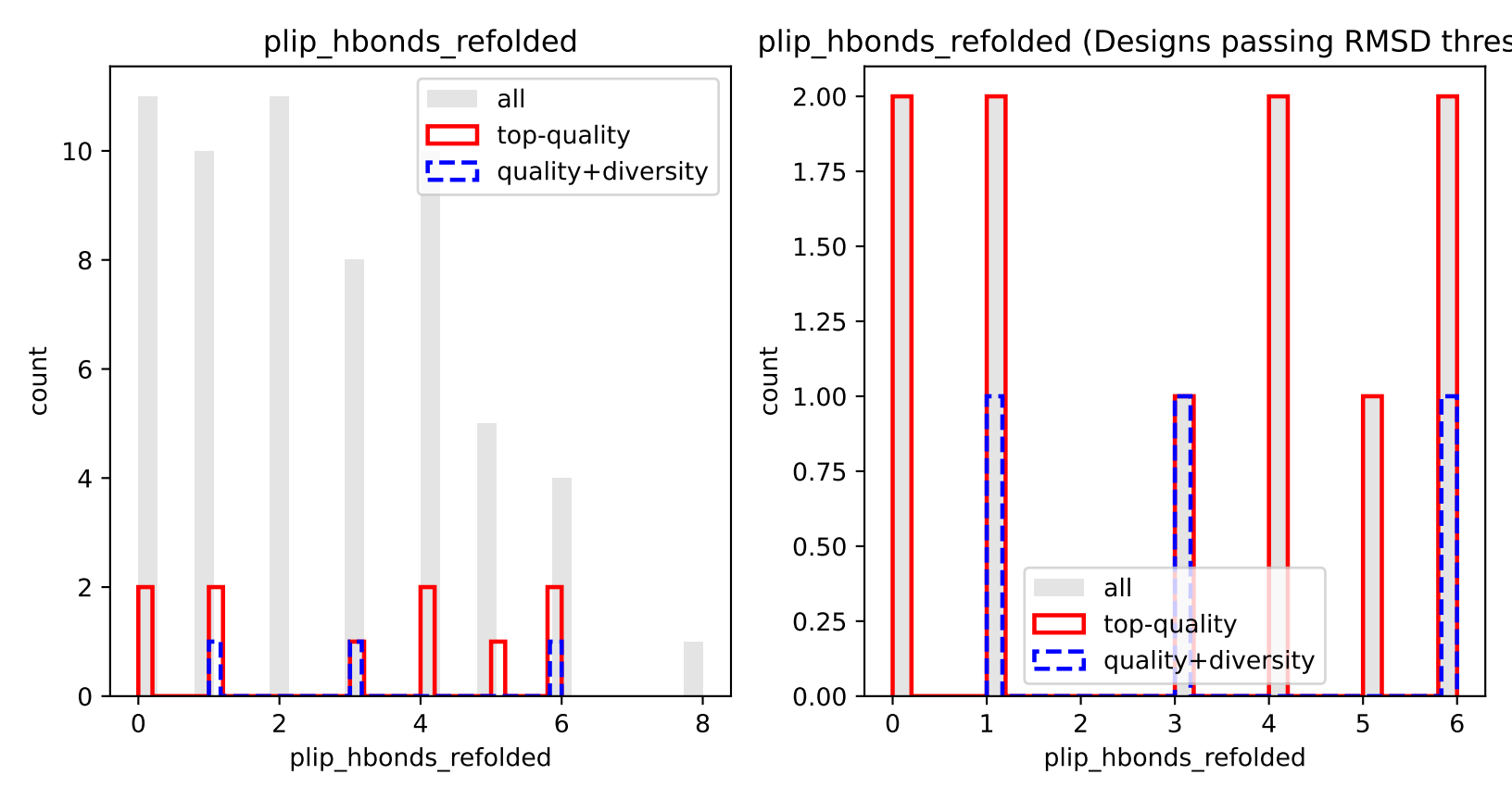

design_iptm
all
top-quality
quality+diversity
count
design_iptm
design_iptm (Designs passing RMSD threshold
all
top-quality
quality+diversity
count
design_iptm

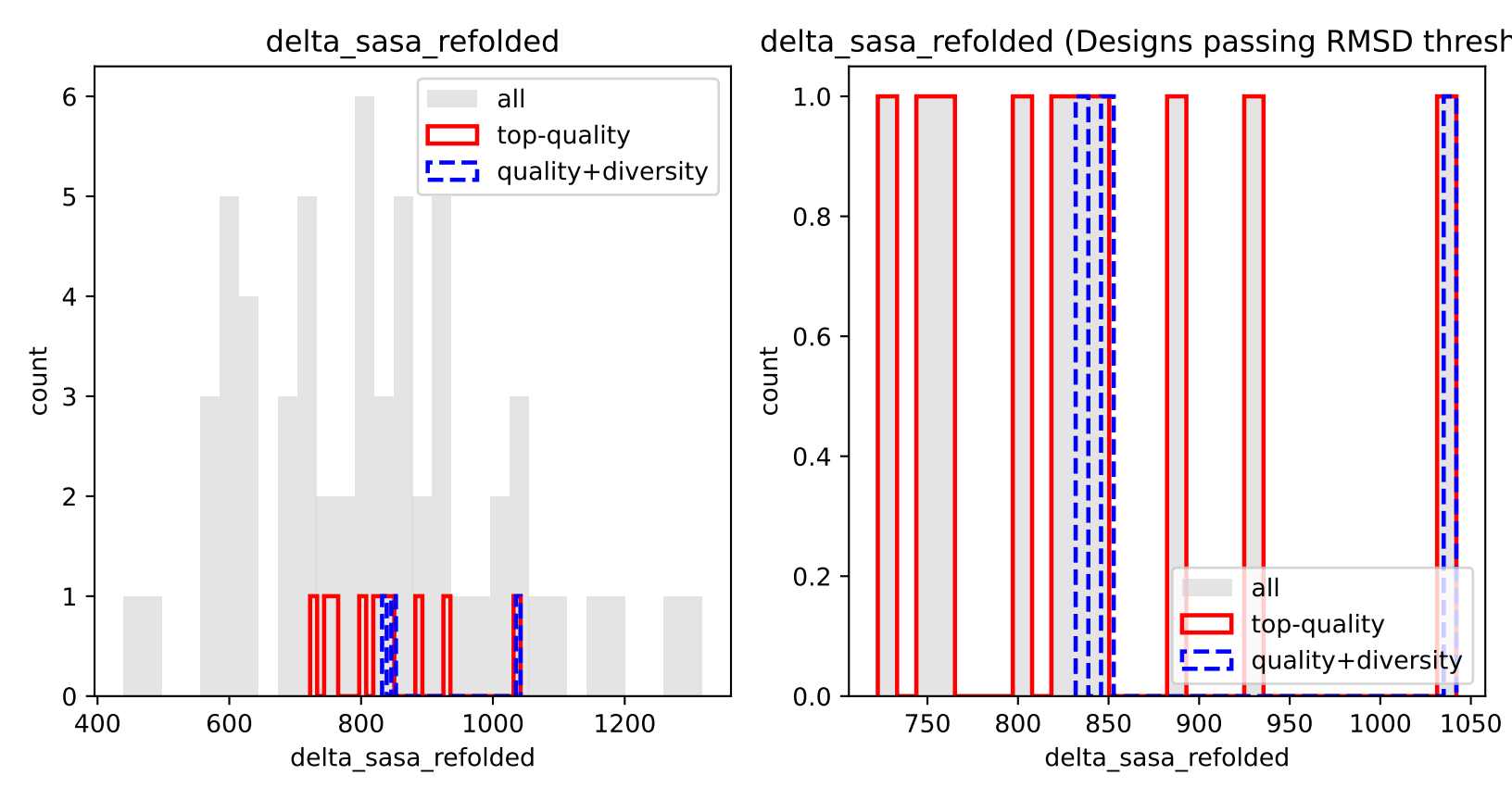

design_ipsae_min
all
top-quality
quality+diversity
count
design_ipsae_min
0.000 0.025 0.050 0.075 0.100 0.125 0.150 0.175
0 5 10 15 20 25 30 35
design_ipsae_min (Designs passing RMSD thresh
all
top-quality
quality+diversity
count
design_ipsae_min
0.0 0.5 1.0 1.5 2.0 2.5 3.0

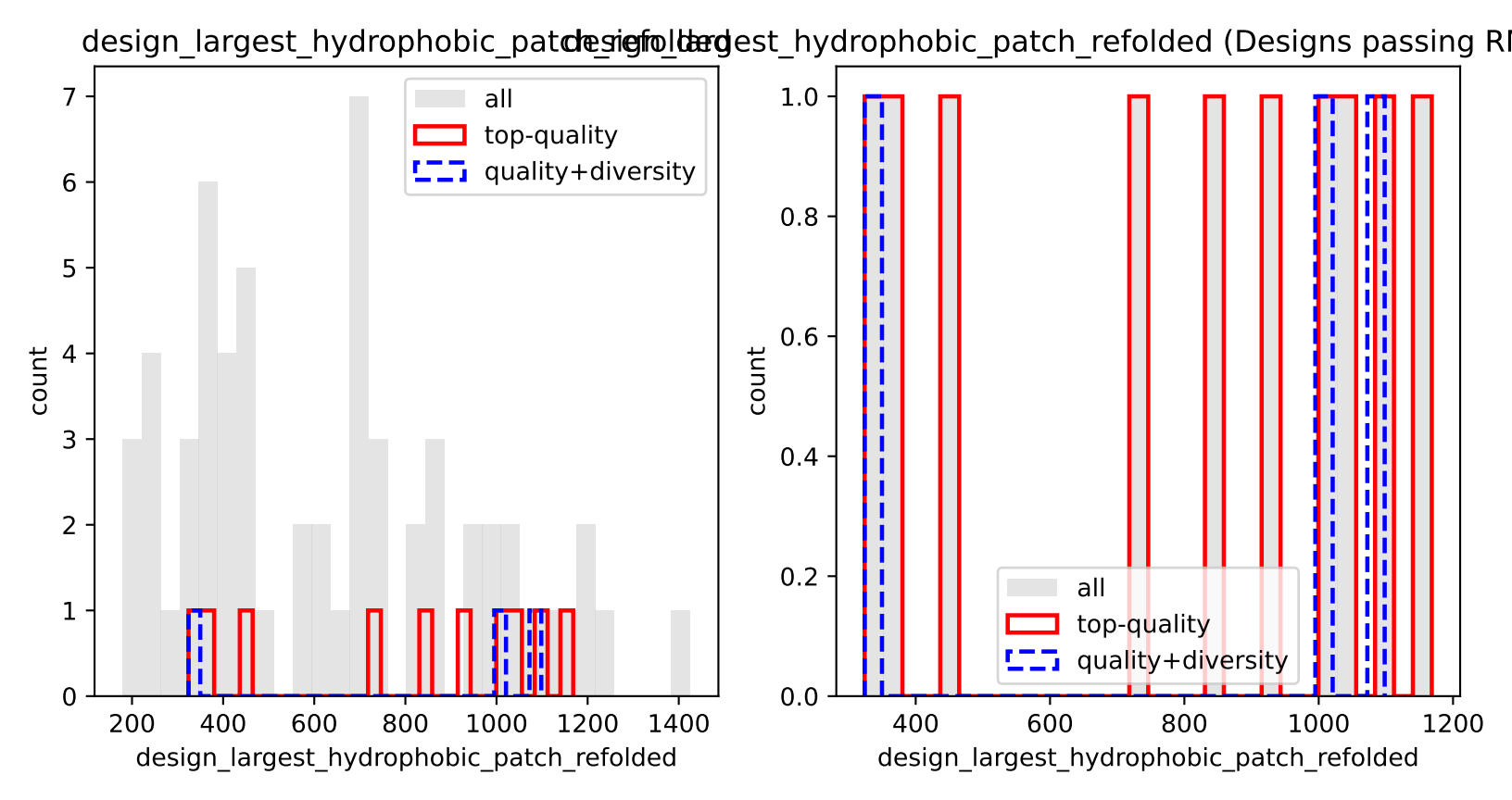

design_to_target_ipsae
all
top-quality
quality+diversity
count
design_to_target_ipsae
design_to_target_ipsae (Designs passing RMSD thre
all
top-quality
quality+diversity
count
design_to_target_ipsae

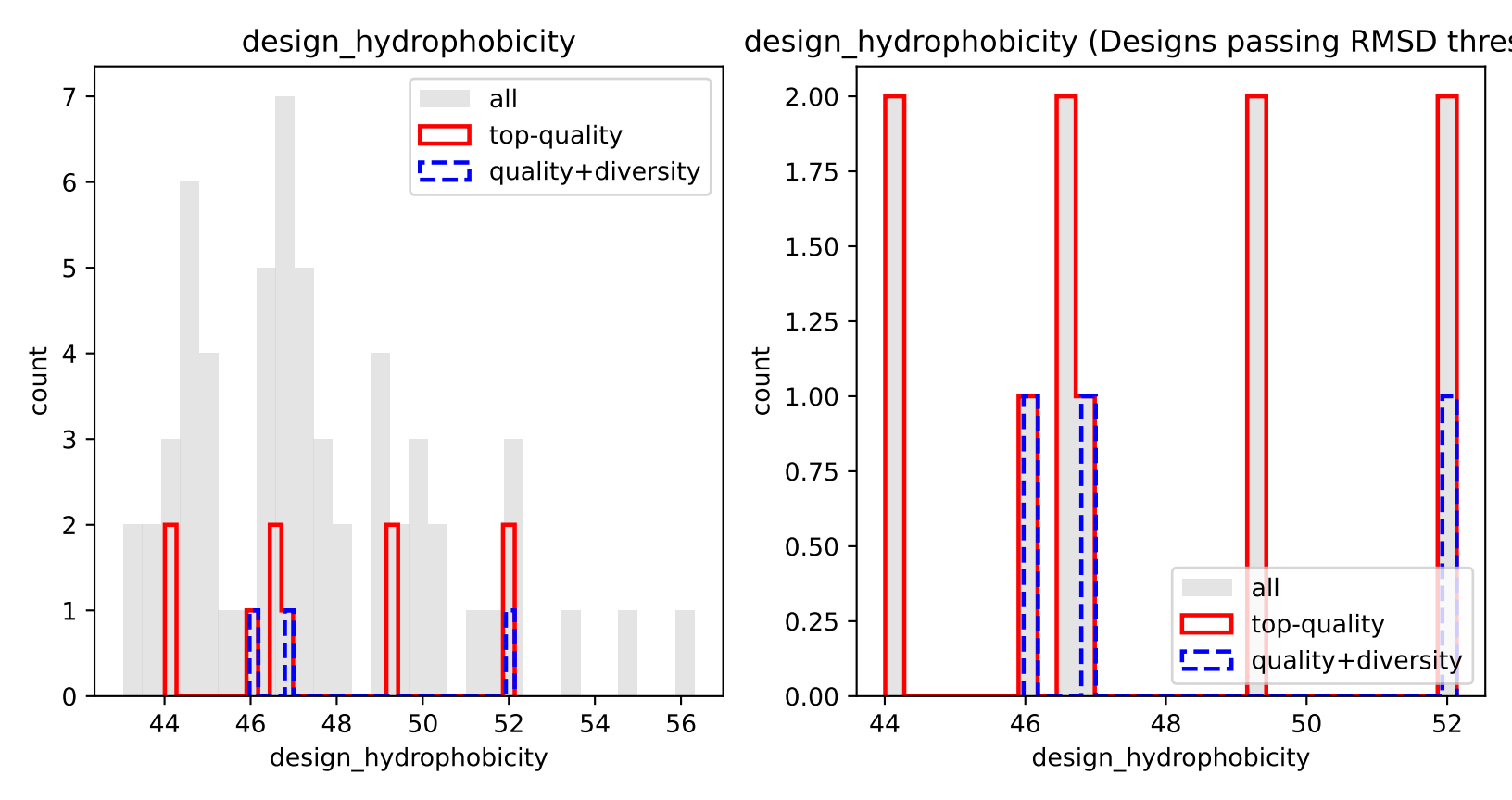

min_design_to_target_pae
min_design_to_target_pae (Designs passing RMSD thr
all
top-quality
quality+diversity
count
min_design_to_target_pae
0
1
2
3
4
5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
5
10
15
20
6
8
10
12
14
16


plip_saltbridge_refolded
all
top-quality
quality+diversity
count
plip_saltbridge_refolded
plip_saltbridge_refolded (Designs passing RMSD thre
all
top-quality
quality+diversity
count
plip_saltbridge_refolded


plip_hbonds_refolded
all
top-quality
quality+diversity
count
plip_hbonds_refolded
plip_hbonds_refolded (Designs passing RMSD thres
all
top-quality
quality+diversity
count
plip_hbonds_refolded


delta_sasa_refolded
all
top-quality
quality+diversity
count
delta_sasa_refolded
delta_sasa_refolded (Designs passing RMSD thresh
all
top-quality
quality+diversity
count
delta_sasa_refolded


design_largest_hydrophobic_patch_refolded
all
top-quality
quality+diversity
count
design_largest_hydrophobic_patch_refolded
design_largest_hydrophobic_patch_refolded (Designs passing RM
all
top-quality
quality+diversity
count
design_largest_hydrophobic_patch_refolded


design_hydrophobicity
all
top-quality
quality+diversity
count
design_hydrophobicity
design_hydrophobicity (Designs passing RMSD thres
all
top-quality
quality+diversity
count
design_hydrophobicity